



280 Ah Liquid Cooling Rack

User's Manual

产品名称: O852280-E/ O852280-P

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- CATL Confidential -

Revision	Chapter	Description	Date
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V1.1	Offloading	Precautions for forklift handing	2021/11/05
V1.2	Charging of Battery Rack	Add the precharge logic	2022/1/10
V1.3	All	Modify	2022/06/15
V1.4	Important Safety Instructions	1. Add LOTO lock description 2. Update the description for shock hazard	2022/07/07
V1.5	Clause 9.2 Battery system inspection	Revise the rack Color from RAL7042 to RAL7035	2022/07/25
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V1.8	Clause 5.1 Site and environment requirements	Modify the site arrangement	2023/03/22

1 Important Safety Instructions

SAVE THESE IMPORTANT SAFETY INSTRUCTIONS

EnerOne battery rack installation and service require knowledge of high voltage electricity and should only be performed by qualified person. CATL assume no liability for injury or property damage due to repairs attempted by unqualified individuals or a failure to properly follow these instructions. These warnings and cautions must be followed when using EnerOne product.

1.1 Electrically Safe Work Condition

A state in which an electrical conductor or circuit part has been disconnected from BESS energized parts, the insulation switches must in "OFF" state. If necessary, temporarily grounded for personnel protection.

1.2 Qualified person

One who has demonstrated skills and knowledge related to the BESS construction, operation, installations and has received safety training to identify the hazard and reduce associated risk.

Qualified persons shall be trained and knowledgeable in the operation of battery system, and specific work methods and shall be trained to avoid the electrical hazards that are present. Such persons shall be familiar with the proper use of precautionary techniques and PPE. Training for a qualified person shall include the following:

1. Skills and techniques to avoid a shock hazard:
 - a. Between exposed energized surfaces, which might include temporarily insulating or guarding parts to permit the employee to work on exposed energized parts.
 - b. Between exposed energized surfaces and grounded equipment, other grounded objects, or the earth itself, that might include temporarily insulating or guarding parts to permit the employee to work on exposed energized parts.
2. Method of determining the battery system working boundaries.

1.3 Safety symbol

Below some basic safety symbols and descriptions are applicable to all equipment on the BESS plant.

	DANGER
	When energized, this equipment presents a potential hazard of electric shock, burn or death. Only authorized personnel who are thoroughly familiar with the equipment and adequately trained may install, operate or maintain this equipment. To prevent death, personal injury, or damage to equipment, follow all site safety procedures as indicated by HSE Plan. To minimize hazard of electrical shock, death and burns, approved grounding practices must be strictly followed.

	WARNING
	To prevent personal injury or damage to equipment, personnel shall adhere to site protocol regarding working at heights。 To prevent personal injury or equipment damage by equipment malfunction, only adequately trained personnel should modify any programmable machine. Always ensure applicable standards and regulations are followed and only properly certified equipment is used as a critical component of the safety system. Never assume that critical-safety control loop is closed.

	WARNING — Shock Hazard
	Do not touch the system connectors or terminals. PPE and related training as required by local codes and regulations are part of the maintenance plan.

	WARNING — Arc Flash Hazard
	There is an arc flash hazard associated with all electrical equipment. There is an increased risk of arc flash with any equipment modification (e.g., open doors). Serious injuries can occur in arc flash incidents. Appropriate PPE and training as per local codes and regulations is required.

	WARNING — Fire Hazard
	Fire is possible under certain, extreme fault conditions.

	CAUTION Indicates a procedure, condition, or statement that, if not strictly observed, could result in damage to or destruction of equipment.
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The following directives must be followed when working with or near the BESS equipment.

1. Safety parameters and procedures are site-specific, therefore, must be developed by the customer.
2. Review and refer to all safety warnings contained in this manual before beginning operations.
Follow all the safety procedures as prescribed by the supplier manufacturer published safety procedures for each equipment.
3. Establish a clear, permanent, restricted access around the system. Fire extinguishers must be readily available on-site.
4. The system requires a clean, mechanically stable operating environment, free from conductive contaminants, combustibles, moisture, liquids and gases and corrosive substances. The system may be operated only when the ambient temperature is between **-25°C to 55°C**. Do not tap, drill puncture or attach anything to the equipment enclosures. Ensure there is enough clearance on all sides of the enclosures to allow access to the system. Remove watches, jewellery, rings and other metallic objects and always use tools with insulated handles.
5. With power applied to the equipment, hazardous voltages will be present on some components. To prevent accidental death or injury, do not touch any components within the enclosure unless specifically directed to do so. To reduce risk of electrical shock, verify that all equipment is properly grounded.
6. Strokes and alarms within the system for evacuation are activated by the FSS system.

	WARNING Only authorized, adequately trained Energy Storage personnel with proper Personal Protective Equipment (PPE) can access the system. Do not open the equipment access doors unless qualified person.
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1.4 Primary Risks Associated With BESS

The primary risks associated with the BESS system are:

1.4.1 Hazardous Voltage

Hazardous Voltage
• High AC at MV Switchgear, Isolation transformer and Auxiliary Transformer • High AC at PCS input, aux transformer input. • 1500 VDC at PCS output, Battery rack.
Consequence
• Electrocution, Fire, Explosion
Avoidance
• Only qualified and authorized person should only be allowed restricted access. • Equipment barriers must remain in place. • PPE must be appropriate for the task and equipment

1.4.2 Arc fault

Arc fault
• When insulation breakdown or accidental contact between high voltage conductors, an arc fault will be happen through the air gap between conductors.
Consequence
• Arc fault creates an electrical explosion. • Light and heat emitted from the explosion is known as <u>arc flash</u>. • Pressure wave caused by the tremendous temperatures of the arc flash is known as <u>arc blast</u>.
Avoidance
• Only qualified and authorized person should only be allowed restricted access. • Restricted access • Equipment barriers must remain in place • PPE must be appropriate for the task and equipment

1.5 Arc Flash hazard boundary

The US National Fire Protection Association (NFPA) has developed specific approach boundaries designed to protect employees while working on or near energized equipment. These boundaries are:

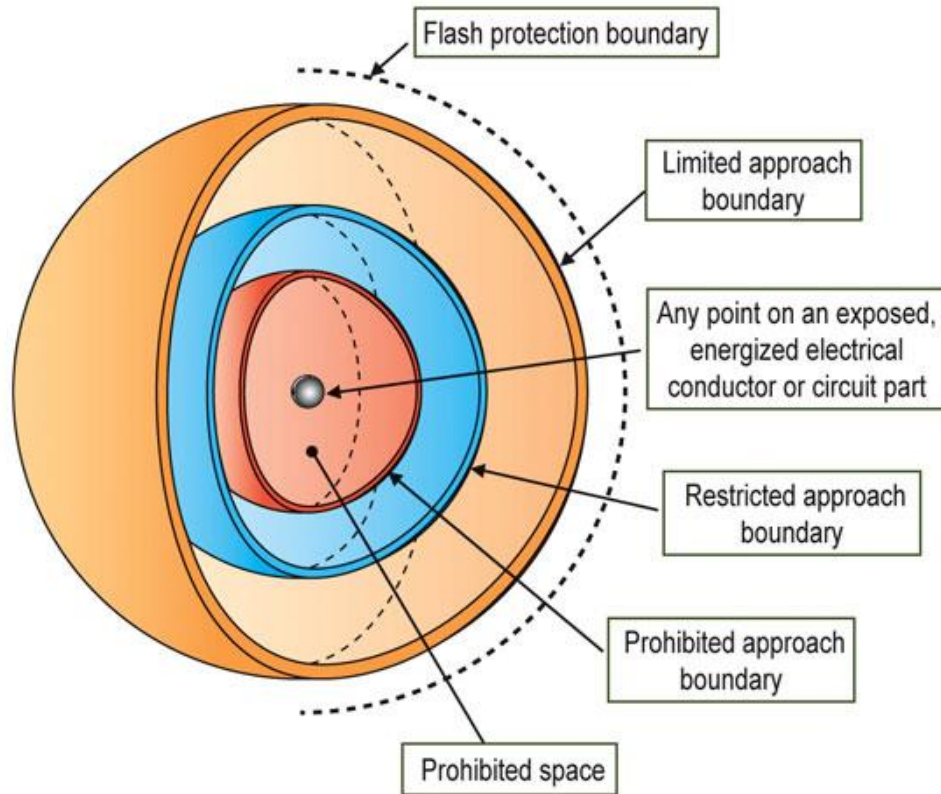


Figure 1-1: Arc Flash Boundaries (Refer to NFPA 70E)

- **Flash Protection boundary (outer boundary)** – this is the farthest established boundary from the energy source. If an arc flash occurred, this boundary is where an employee exposed to a curable second-degree burn. The issue here is the heat generated from a flash that results in burn.
- **Limited approach** – An approach limit at a distance from an exposed live part where a shock hazard exists.
- **Restricted approach** – An approach limit at a distance from an exposed live part which there is an increased risk of shock.
- **Prohibited approach (inner boundary)** – A distance from an exposed part which is considered the same as making contact with the live part.

1.6 Arc Flash Label

According to "Maximum Power Method" in NFPA 70E, the estimated DC arc flash incident energy at the maximum power point can be calculated, and then the hazard boundary is defined as following Figure 1-2.

 WARNING	
Arc Flash and Shock Hazard Appropriate PPE Required	
0' -11" 0.95 #1	Flash Hazard Boundary cal/cm2 Flash Hazard at 1.5 inches PPE Level Cotton underwear plus FR shirt & FR pants plus FR coverall
1.5 5' - 0" 1' -8"	kV Shock Hazard when cover is removed Limited Approach Restricted Approach
Equipment Name: EnerOne Battery Rack VALID FOR NORMAL SYSTEM CONFIGURATION ONLY.	

Figure 1-2: Arc Flash Label

1.7 Personal Protective Equipment (PPE)

The PPE is determined by the incident energy which is the temperature produced (in cal/cm²) at a distance (usually eighteen inches) from an arc flash.

PPE requirement for EnerOne battery rack operational precautions is category 1

PPE CATEGORY 1	PPE CATEGORY 2	PPE CATEGORY 3	PPE CATEGORY 4
Minimum Arc Rating of 4 cal/cm²	Minimum Arc Rating of 8 cal/cm²	Minimum Arc Rating of 25 cal/cm²	Minimum Arc Rating of 40 cal/cm²
Arc Rated Clothing: • AR long-sleeve shirt and pants, or AR coverall • AR face shield, or AR flash suit hood • AR jacket, parka, rainwear, or hard hat liner (as needed)	Arc Rated Clothing: • AR long-sleeve shirt and pants, or AR coverall • AR flash suit hood, or AR face shield and AR balaclava • AR jacket, parka, rainwear, or hard hat liner (as needed)	Arc Rated Clothing: • As required: AR long-sleeve shirt, AR pants, AR coverall, AR flash suit jacket, and/or AR flash suit pants • AR flash suit hood • AR gloves • AR jacket, parka, rainwear, or hard hat liner (as needed)	Arc Rated Clothing: • As required: AR long-sleeve shirt, AR pants, AR coverall, AR flash suit jacket, and/or AR flash suit pants • AR flash suit hood • AR gloves • AR jacket, parka, rainwear, or hard hat liner (as needed)
Protective Equipment: • Hard hat • Safety glasses or safety goggles • Hearing protection (with inserts) • Heavy-duty leather gloves • Leather footwear (as needed)	Protective Equipment: • Hard hat • Safety glasses or safety goggles • Hearing protection (with inserts) • Heavy-duty leather gloves • Leather footwear	Protective Equipment: • Hard hat • Safety glasses or safety goggles • Hearing protection (with inserts) • Leather footwear (as needed)	Protective Equipment: • Hard hat • Safety glasses or safety goggles • Hearing protection (with inserts) • Leather footwear (as needed)

Figure 1-3: PPE Category

1.8 LOTO Procedure

1. The battery racks, which are parallel connected to one block, needs to be powered off first. Check the status of PCS and battery racks from EMS, make sure that the circuit breakers/contactors of PCS DC side are all disconnected, and the HV contactors of all racks are disconnected.
2. PPE 1 is required for operator.
3. Turn off the on-site AC circuit breaker which supplies the external auxiliary power of all racks. Apply a LOTO adapter and LOTO lock and tag to the circuit breaker handle to prevent it from being turned back on.
4. Switch off the isolation switch of the rack to "OFF" state. Pull out the LOTO ring of the isolation switch handle and apply LOTO lock and tag to the LOTO ring to prevent it from being turned back on.
5. Turn off the AC circuit breaker QF1&QF2 inside the battery rack that connect the external auxiliary power.
6. Check, with a multi-meter, that the 220 VAC auxiliary power input connection terminal JXH1 reads 0 V.
7. Check, with a multi-meter, that the 1,500 VDC bus bar reads 0 V.
8. Install a safety ground clamp from the positive 1,500 VDC bus bar to the ground bar. Use a shotgun stick to make the initial contact between the positive 1,500 VDC bus bar and ground.
9. Install a safety ground clamp from the negative 1,500 VDC bus bar to the ground bar. Use a shotgun stick to make initial contact between the negative 1,500 VDC bus bar and ground.

2 Manual Overview

2.1 Applicable Scope

This manual is applicable to the EnerOne battery rack.

All references to “battery rack” in this manual, unless specifically indicated here in, refer to the product series mentioned above.

2.2 General

This manual has been prepared primarily for Battery Energy Storage System (BESS) operators and service personnel and provides three essentials: operation, maintenance and parts information for the BESS. A distribution of text revisions when necessary will be made to all recipients of the Operations and Maintenance manual.

Personnel who practice good operating and maintenance procedures ensure BESS reliability. It is of vital importance that station personnel, trained or untrained in BESS operation, become thoroughly familiar with the information contained in this manual. Prior to BESS operation, a sound program of maintenance should be developed and instituted. Adherence to such a program will result in reduced operating costs, fewer outages, and shorter periods of downtime.

2.3 Manual Updates

An experienced staff from the best information available has compiled the manuals provided for this BESS equipment prior to the time of shipment. Due to your shipping requirements and our desire to meet manual delivery commitments, manuals may be shipped with missing information. All manuals in your hands should be updated as material is received from CATL Company. To assist you in revising and updating your manuals, we offer the following suggestions:

1. Identify those individuals or functions who will be using the sets of manuals.
2. Assign the manuals to each individual or function and number each set.
3. Maintain a log of manual sets and the associated owners.
4. If possible, assign a librarian to ensure all controlled (or numbered) sets of manuals are updated as material is received from CATL. If this is not possible, each owner must keep their set of manuals current.

5. Retain shortage/revision instructions sheet at the front of the appropriate volume. This acts as a log of updates received.
6. Follow instructions for insertion of new material exactly. If the material is not filed properly, it cannot be easily found.
7. Insert shortage/revision material in manuals as soon as received. Some updates are small and can be misplaced.
8. All other manuals left from installations, etc., should be clearly labelled as extras, spares, or not controlled copies.

2.4 Abbreviation

CATL	Contemporary Amperex Technology Limited
BMU	Battery Management Unit
BMS	Battery Management System
BESS	Battery Energy Storage System
EMS	Energy Management System
PCS	Power Conversion System
CSC	Cell Supervision Circuit
CSU	Current Sample Unit
ETH	Ethernet Module
FSS	Fire Suppression System
HV	High Voltage
LV	Low Voltage
MSD	Manual Switch Disconnecter
MBMU	Master battery management unit
SBMU	Slave Battery Management Unit
CAN	Controller Area Network
ACAN	CAN between BMU and HCU
CCAN	AN between BMU and CSC
SCAN	CAN between BMU and CSU
DCAN	CAN for calibration
BOL	Begin of Life
EOL	End of Life
SOC	State of Charge
SOE	State Of Energy
SOH	State of Health
SOP	State Of Power
TMS	Thermal Management System

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3 Product Introduction

3.1 Product Overview

EnerOne Battery rack is the core unit in the energy storage system and acts as the equipment for storing electrical energy. It can be applied to many applications including renewable energy integration, frequency regulation, and voltage regulation of the grid.

This battery rack consists of 1P52*8S battery modules system, battery management system (BMS), fire suppression system (FSS), thermal management system (TMS).

Table 3-1: Components of battery rack

System	Sub Components	Quantity	Remark
Battery Racks	Battery Modules	8	With CSC
BMS	Control box	1	Including SBMU, fuse, isolation switch and so on
TMS	Cooling unit	1	Including compressor, pump, fan, heater and others
FSS	Smoke detector	1	Type can be configured by request
	Heat detector	1	Type can be configured by request
	Aerosol	1	Type can be configured by request



Figure 3-1: EnerOne Outdoor Liquid Cooling Rack

3.1.1 Battery Rack

The battery system contains 8 battery modules by series connection, each battery module is composed of 52 battery cells in series connection also, so each rack contains 416 battery cells. Totally, EnerOne liquid-cooled rack's configuration is 1P416S.

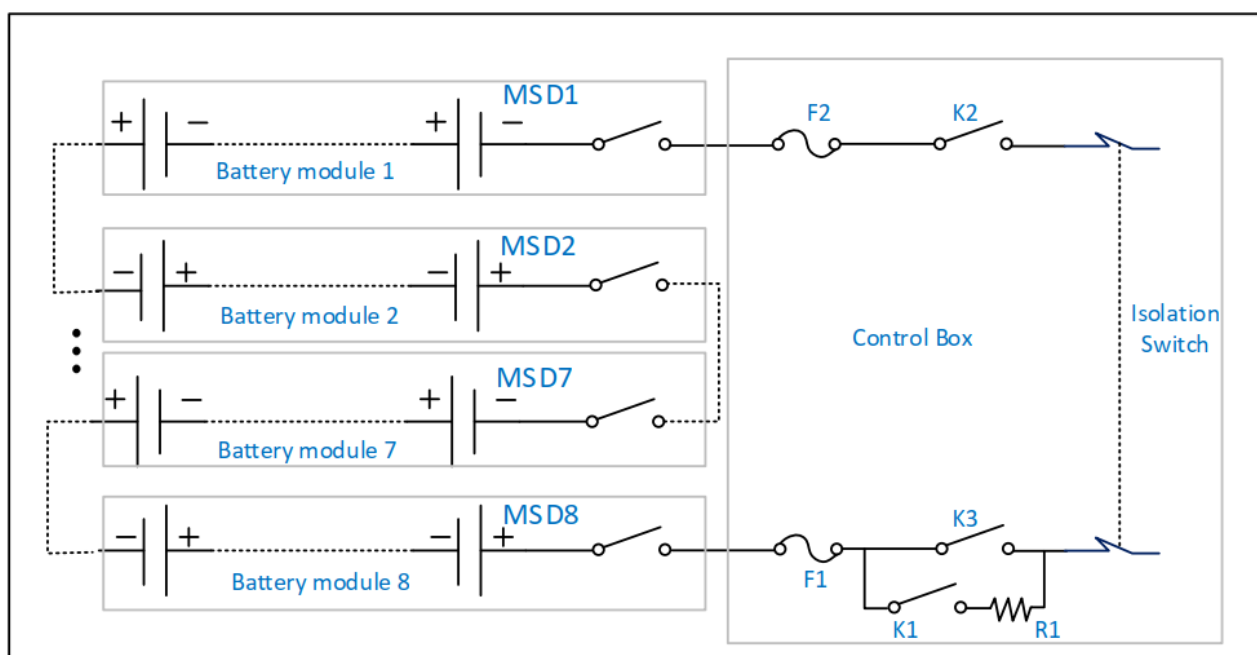


Figure 3-2: Circuit diagram of battery rack

3.1.2 Battery Module

Total 52 pieces lithium iron cells (280Ah/3.2V) in series connection is used for every battery module. For safety protection, an internal high speed DC fuse is included, and removable MSD switch can cut off the high voltage connection during transportation process.



Figure 3-3: Liquid cooling battery module

3.1.3 Control Box

Control box integrate SBMU, HV contactors, pre-charge circuit, high breaking capacity and high speed DC fuses, which can control battery rack by SBMU and send battery status data to MBMU via CAN bus. Detailed interface shown in Figure 3-4,

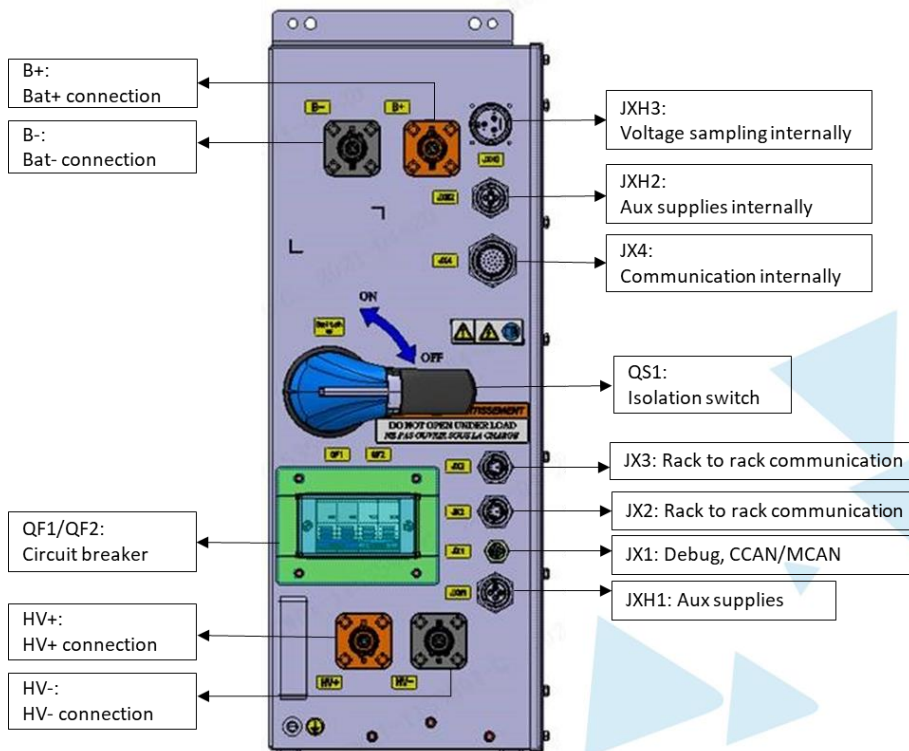


Figure 3-4: Control box

Table 3-2: Information of connector

No.	Connector	Definition	Connector information	Remark
1	JX1-a	M-CAN-H	Plug: PT06E-8-4S(005)(Soldered) Adapter cable diameter: AWG 18~20 Manufacturer: Amphenol	Debug for commissioning
2	JX1-b	M-CAN-L		
3	JX1-c	C-CAN-H		
4	JX1-d	C-CAN-L		
3	JX2-a	M-CAN-H	Plug: LPT06SE-12-8S(005)(LC)(Crimped) Plug: PT06E-12-8SW(005)(Soldered) Adapter cable diameter: AWG 18~20 Manufacturer: Amphenol	Connect to the next control box
4	JX2-b	M-CAN-L		
	JX2-c	Smoke detector output+		
	JX2-d	Smoke detector output-		
	JX2-e	Heat detector output+		
	JX2-f	Heat detector output-		
	JX2-g	Aerosol output+		
	JX2-h	Aerosol output-		
5	JX3-a	M-CAN-H	Plug: LPT06SE-12-8SW(005)(LC)(Crimped) Plug: PT06E-12-8S(005)(Soldered) Adapter cable diameter: AWG 18~20 Manufacturer: Amphenol	Connect to the previous control box
	JX3-b	M-CAN-L		
	JX3-c	Smoke detector output+		
	JX3-d	Smoke detector output-		
	JX3-e	Heat detector output+		
6	JX3-f	Heat detector output-		
7	JX3-g	Aerosol output+		
8	JX3-h	Aerosol output-		
10	JXH1-a	Power Input L	Plug: LPT06E-14-AAP(005)(LC)(Crimped) Plug: PT06E-14-AAP(005)(Soldered) Adapter cable diameter: AWG 12 & 4mm ² Manufacturer: Amphenol	Auxiliary power supply for BMS
11	JXH1-b	Power Input N		Auxiliary power supply for Chiller
12	JXH1-c	Power Input L		
13	JXH1-d	Power Input N	Plug: ES095-03C95-2SYZ-01 (orange) Adapter cable diameter: AWG 3/0 & 95mm ² , Unshielded Cable Manufacturer: SANCO	Connect to the next rack
14	HV+	Positive output		
15	HV-	Negative output		

No.	Connector	Definition	Connector information	Remark
			Adapter cable diameter: AWG 3/0 & 95mm ² , Unshielded Cable Manufacturer: SANCO	
16	Bat+	Positive input	Plug: ES095-03C95-2SYZ-01 (orange) Manufacturer: SANCO	Connect to battery module
17	Bat-	Negative input	Plug: ES095-03C95-1SYW-01 (black) Manufacturer: SANCO	

Table 3-3: Information of switch

No.	Switch	Name of switch	Function
1	QF1/QF2	Circuit Breaker	To switch on and off the DC24V auxiliary power supply of the control box
2	QS1	Isolation switch	To switch on and off the output of battery system. (Cannot operate with load)

3.1.4 FSS overview

The FSS is composed of smoke detector, heat detector and aerosol, which main function is to prevent fire spread in time when any open flame signal or gas signal appears in the battery system and sent out fire signal to EMS system. The main components layout is shown as Figure 3-5.

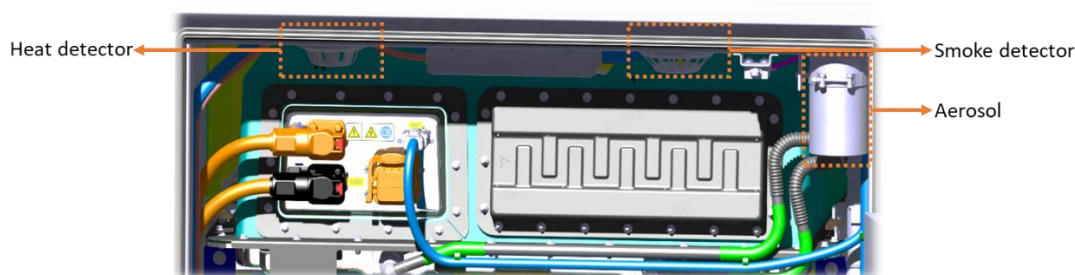


Figure 3-5: FSS system

Table 3-4: Main components of FSS system

Component	Quantity	location
Smoke detector	1	On the top of Battery rack
Heat detector	1	On the top of Battery rack
Aerosol	1	On the top of Battery rack

3.1.4.1 FSS Control Logic Overview

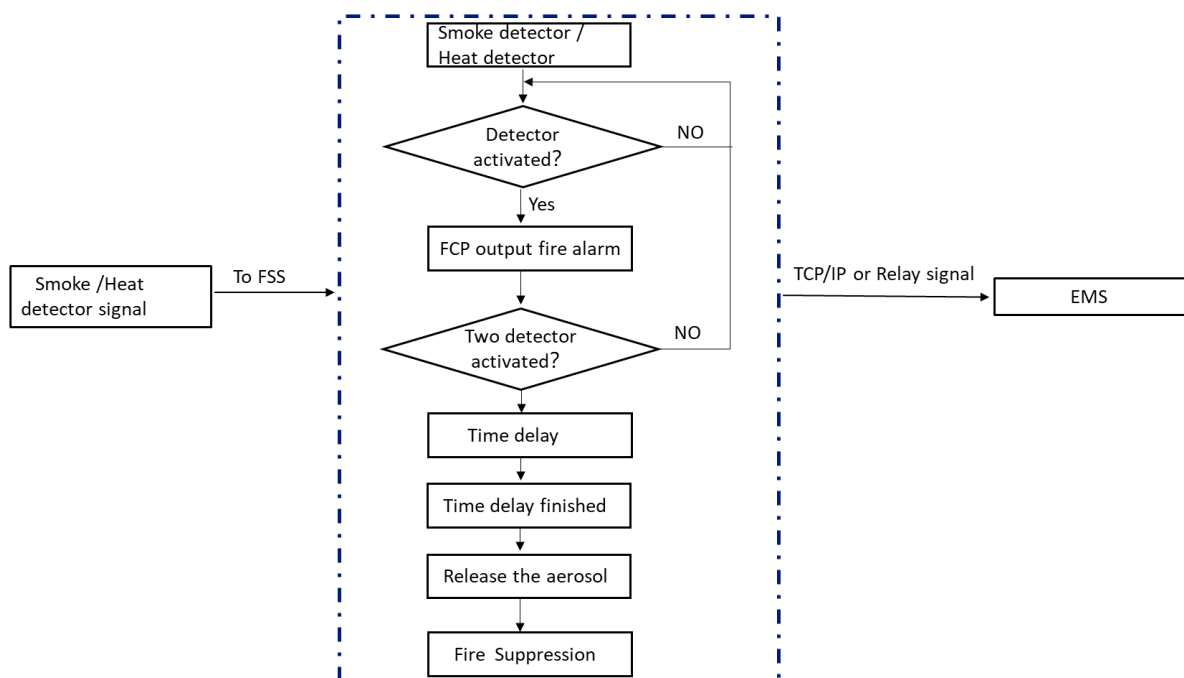


Figure 3-6: FSS Control Logic

3.1.5.3 Instruction of Smoke Detector, Heat Detector and Aerosol

Table 3-5: Instruction of Smoke Detector, Heat Detector and Aerosol

FSS components	Image	Function
Smoke detector		Working principle: using a light source and a photosensitive sensor, when smoke particles enter the light path, some of the light is scattered by reflection and refraction onto the sensor.
Heat detector		Working principle: to detect fire by using heat sensitive elements.
Aerosol		Working principle: Aerosol mainly composed of ultra-fine potassium salt particles and inert gas, and ultrafine potassium salt particles can quickly consume OH, H and O radicals and prevent the combustion chain reaction from proceeding.

3.1.5 TMS Overview

The TMS is liquid cooling, whose main function is to maintain the operating temperature of the battery

system to an allowable range. The chiller is used for providing cooling & heating capacity for each battery rack. Each module has their own outlet and inlet pipes, and controlled by TMS controllers.

Due to adapt high performance TMS, the whole system 416 cells can acquire a high consistency and then can keep a long service time. The main feature of this system included:

1. Refrigerant type: R134a
2. Suitable for extreme ambient temperature application: -25°C... +55 °C
3. Airflow requirement: $\geq 5500\text{m}^3/\text{h}$
4. Single Phase L-N 50/60Hz power supply are prepared by users through JXH1 connector
5. The chiller is included for one rack, and 3kW cooling power can be acquired for 0.5P system, 8kW cooling power can be acquired for 1P system.

3.2 Product Specification

Table 3-6: Battery Rack Characteristics

No.	Item	Specification	Remark
1	Configuration	1P416S	
2	Weight	3500±300kg	
3	Dimension	1300mm(W)*1300mm(D)*2340mm(H)	
4	Nominal Capacity (kWh)	372.7kWh	
5	Voltage Range (Vdc)	1164.8~1497.6 Vdc	
6	Nominal Voltage	1331.2 Vdc	

3.3 BMS Introduction

3.3.1 BMS Overview

BMS is used in conjunction with the ESS energy storage system, which can monitor the battery voltage, current, temperature, managing energy absorption and release, thermal management, low voltage power supply, high voltage security monitoring, fault diagnosis and management, external communication with PCS and EMS, ensure the stable operation of the energy storage system.

The BMS system is composed of 1 unit of MBMU, 1 unit of ETH, 1 units of SBMUs, 8 units of CSCs.

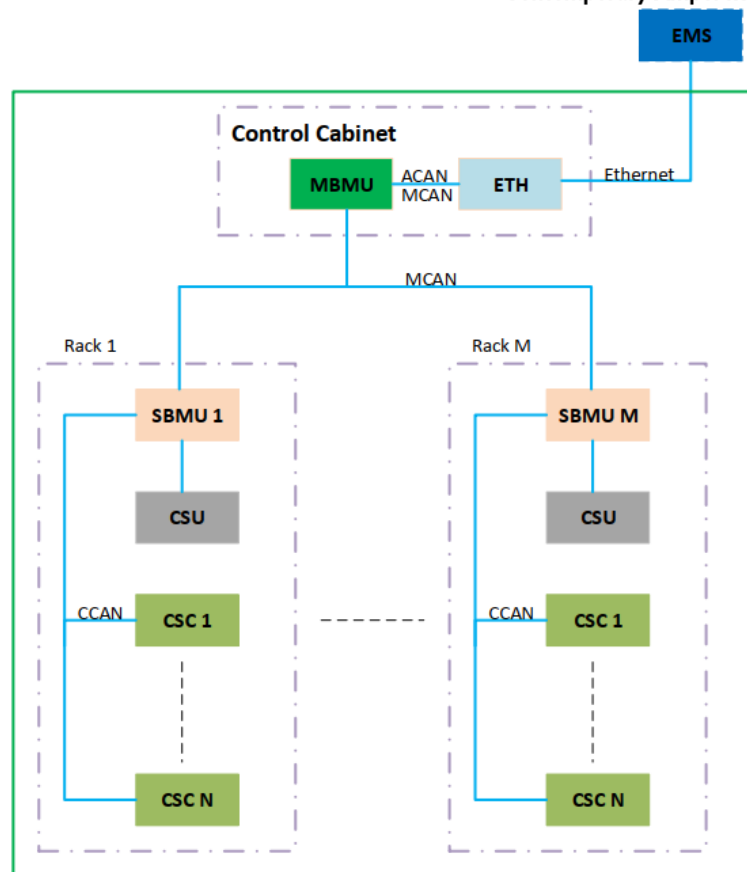


Figure 3-7: Communication Topology

In Figure 3-7, M represents quantities of rack, in order to ensure the quality of communication, $M \leq 20$.

3.3.2 Battery Status Monitoring

1. BMS monitors the battery's parameters, including cell voltage, module temperature, battery current and total battery voltage.
2. BMS detects the battery status such as State of Charge (SOC), SOH and the calculation of SOP.
3. BMS functions as a safety management system in such cases as under voltage, over discharge, overvoltage, over-temperature, and overcurrent of the battery. In case of failure, the system will give an alarm to the supervisory equipment, limit the charge and discharge current or power, and delay the disconnection of all HV relays. This can protect the battery while safeguarding the power systems security.
4. BMS shall provide battery information (incl. data recording and fault waveform recording) to EMS.

3.3.3 Charging/Discharging Management

1. BMS controls and monitors HV relays, including the contactor's low voltage coils and high-voltage contact (auxiliary contact).
2. BMS has pre-charge control within the parallel connection among racks.

3. BMS works in the management of charge and discharge. It will calculate the charge and discharge power limit according to the existing status of the battery (temperature, SOC) and actual performance of electrical components and then report to EMS which has the function of controlling to these limits;
4. BMS has the function of balance Management to extend the operability of the battery system.

3.3.4 Thermal management

1. BMS has the function of sample collecting of battery cell temperature and chiller operating status.
2. BMS controls the liquid cooled TMS system based on battery cell & coolant's temperature.

3.3.5 Program Refreshing

BMS can flash programs on site, which supports the flashing of MBMU, SBMU, CSC and ETH by using the host computer through MCAN.

3.3.6 High Voltage Safety monitoring

1. BMS has the function of high voltage sampling (collecting data of the main positive voltage)
2. BMS supports the detection of the dry contact of MSD, Fuse and switch, as well as the auxiliary contact of the primary loop relay.

3.3.7 Fault Diagnosis Management

1. BMS stores information such as operational parameters and historical alarms that can be viewed by ESS host computer.
2. BMS enables storage of latest historical alarms.

4 Transportation and Storage

4.1 Transportation process and requirements

Transportation scheme: sea cargo and land carriage.

During loading and unloading, it should be handled with care to prevent throwing, rolling, and heavy pressure; during the transportation, external mechanical impact should be avoided.

- Battery rack transportation requirement for land carriage
 1. Transport scheme for bulky cargo must be evaluated in advance;
 2. The high-speed driving speed of the vehicle (truck) shall not exceed 100 km / hour & domestic legal requirement;
 3. During the driving process, it is forbidden to make sudden braking and sharp turns;

4. Keep the vehicle in good condition, check the vehicle's carrying status frequently, find and solve the problems in a timely manner;
- Battery rack transportation requirement for sea cargo
1. Keep the vehicle in good condition, check the vehicle's carrying status frequently, find and solve the problems in a timely manner.

Requirements for Battery Rack Transport

Based on battery features and with the purpose of maximizing the battery performance, the transport and storage of battery rack should meet the following requirements:

- Allowable transport temperature: $-30^{\circ}\text{C}\sim+60^{\circ}\text{C}$
- Humidity: $\leq 95\%$ (no condensation).
- Take appropriate protection measures to keep SOC level at about 30% and ensure that no short circuit happens, or no liquid is allowed into the battery rack or soaked in liquids (e.g. water, oil, etc.).

4.2 Hoisting & lifting equipment

Equipment requirement

1. According to the site conditions and mechanical equipment, the selection of good performance of lifting equipment, rigging, shackle calculation selection; Ensure that the crane and rope meet the load-bearing requirements.
2. All kinds of sling, including sling, shackle, etc. should be checked in good condition before entering the site; before formal lifting, it should be reconfirmed that the specifications and quality meet the requirements of this lifting.
3. The lifting point is located at the top of the four terminal corners of the rack.
4. When installing or removing a hoisting device, do not drag the rack. Otherwise, the rack may be damaged.
5. Transfer the rack to the designated location by means of forklift transfer or hoisting (priority recommended top 4 points hoisting transfer). The lifting lug should be repainted after the installation.
6. The radius of the hole on the lifting lug is 11mm.

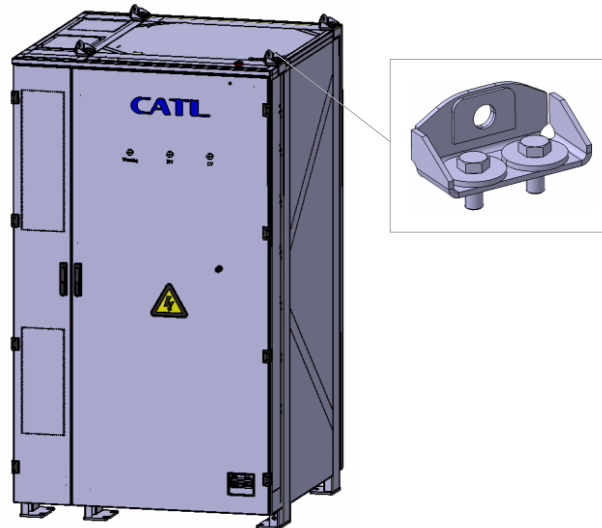


Figure 4-1: Lifting lug on the rack



CAUTION

The gravity center of the battery rack must be kept stable during lifting process.

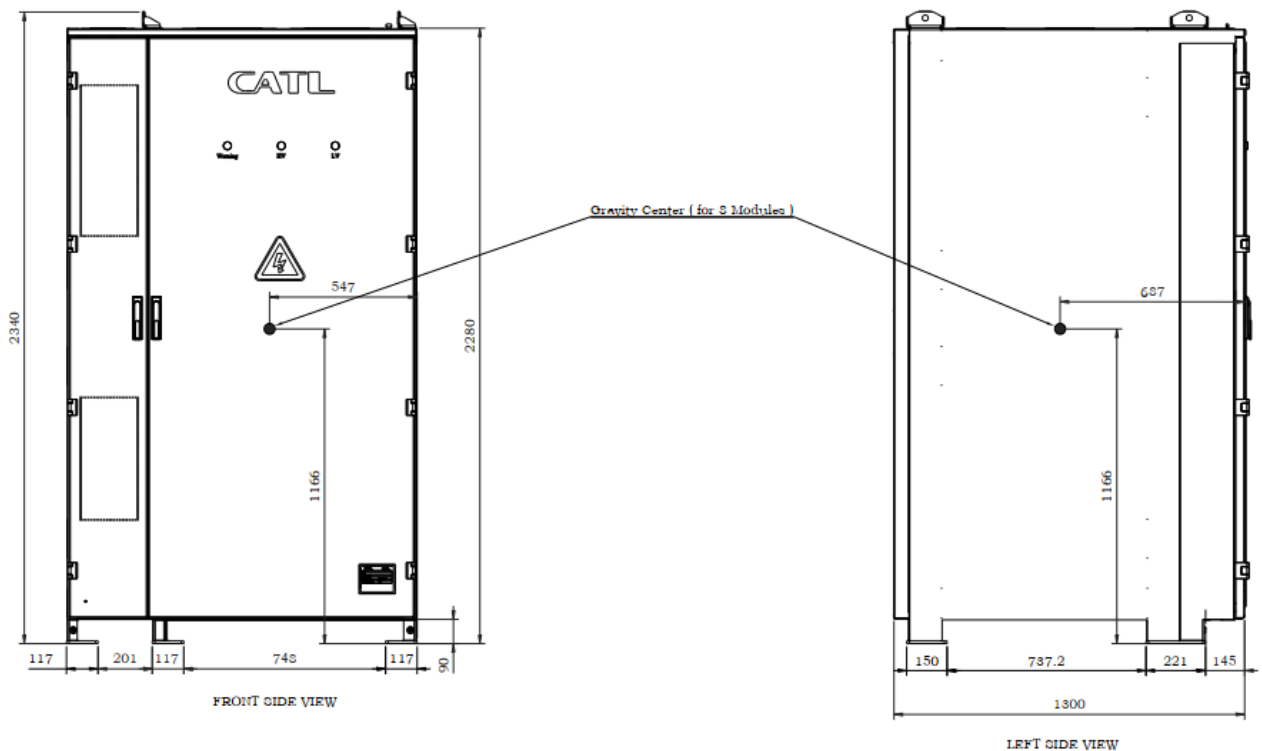


Figure 4-2 Gravity center of battery rack

4.2.1 Precautions for hoisting

Hoisting	Precaution
Before lifting	The lifting capacity and working radius of the crane meet the requirements. If the onsite working conditions do not meet the requirements, seek professional evaluation.
	For outdoor use, it is advised to lift racks when the weather is clear and there is no wind & rain.
	Ensure that the crane and steel cable meet the requirements before hoisting.
	The door of rack is closed and locked.
	Ensure the cable connection is safe and reliable.
During lifting	It is strictly forbidden for irrelevant personnel to enter the lifting area, and it is strictly prohibited to stand under the lifting arm.
	Ensure that the crane is in a proper position, not long distance hoisting.
	Keep the rack steady, and the diagonal inclination of the rack is less than or equal to 5°.
	Hoisting gently, the box body should be slow and steady when falling, to avoid impact on internal equipment.
	When the rack is in contact with the base, remove the steel hoisting cable after the base is evenly stressed.
	Racks can be hoisted only after they are secured.

4.2.2 Risk identification for hoisting

	WARNING
	To prevent personal injury or equipment damage, the risk identification must be done before hoisting

1. The crane in the operation of the human body caused by extrusion or impact;
2. The crane hook overloading fracture, lifting when the sling slip out of the hook;
3. Heavy objects fall in the lifting cause striking, heavy objects fall from the air to the ground and rebound wounding;
4. Accidental contact of spreader or sling with conductive slip line;

5. Truck crane working site ground is not smooth, support unstable, weight imbalance, heavy weight exceeds the rated lifting weight caused by crane overturning;
6. Excessive wind, illegal operation caused the crane overturned;
7. People standing under the boom and other dangerous areas;
8. A person standing or sitting on a hook;
9. When lifting, there is no command, there are people staying in the working area, and the spares and heavy objects of the crane in operation swing and strike pedestrians;
10. The crane and other operators do not wear safety helmet and other personal protective equipment;
11. The driver's cab glass is not cleaned, resulting in unclear vision;
12. The driver has poor communication with the commanding personnel, or misunderstands the hoisting signal;
13. The hanging method is not correct, causing heavy objects to fall out from the hook;
14. The sling used exceeds the safety factor;
15. The wire rope jumps out of the pulley groove;
16. Brake crack, friction gasket wear too much.

4.2.3 Measures of hoisting safety

	WARNING
	To prevent personal injury or equipment damage, the safety measure must be followed

1. Strengthen the management of the construction site, set up a warning area on the hoisting site, send full-time safety personnel to supervise and alert, non-operation personnel are strictly prohibited to enter;
2. Carefully do the preparatory work before lifting, according to the requirements of the plan to prepare machinery and heavy lifting rigging. Strictly implement the lifting equipment performance inspection, lifting rigging inspection, pre-lifting detection and lifting procedures to ensure the safe and reliable lifting operation;
3. Detailed technical disclosure shall be made to all personnel participating in the construction operation in advance, so that they must understand the essentials, procedures and requirements of the

operation. After the crane enters the site, the crane driver will be introduced to the hoisting scheme in detail, and the unified command signal will be made clear.

4. Crane position operation and walking route should be carried out according to the requirements of the scheme, the outrigger pad must be safe and reliable. Special subgrade box cushion is used under the leg to expand the unit force surface. Crane lifting operation should be smooth, slow action, lifting operation should pay close attention to the settlement of the crane leg;
5. The contact between the site commander and the crane driver should be timely and reliable, and the command signal, flag language and gesture should be clear. If abnormal conditions are found, the commander in chief of the lifting site should be reported in time, so as to take effective measures as soon as possible;
6. When the wind reach level 5 or above, it is strictly prohibited to lift in thunder, rain and fog days;
7. Construction personnel entering the hoisting operation site must strictly implement the safety rules and regulations of the site, wear labor protection clothing and safety helmet according to the provisions;
8. Hoisting operation should be unified command, operation personnel should do their duty, closely cooperate, to complete the hoisting operation safely and smoothly.

4.3 Storage

The battery rack should be stored in a place without harmful gases, flammable or explosive products and corrosive chemicals nearby. Keep away from mechanical impact, heavy pressure and strong magnetic field. The distance to heat source should be far away.

4.4 Ambient Thermal Insulation

The battery system must operate within the range of optimum operating temperature, so that the service life of a battery can be extended, and the safety performance of a battery can be improved. The limitation of temperature should completely comply with various definitions in the specification, the installation space of the battery system should be well-ventilated and be designed with the function of thermal insulation.

5 Product Installation

5.1 Site and environment requirements

EnerOne battery rack applies to general outdoor scenarios. In accordance with local laws and regulations, the site selection requirements are as follows:

1. The installation position of the node cannot be in a low-lying area, and the site level is higher than the highest historical water level in the area.
2. The soil condition is good and the ground is solid. No bad geological conditions such as rubber soil and soft soil layer are allowed. The ground that is easy to accumulate water and sink should be avoided.
3. Invest in a well-ventilated place.
4. Keep away from strong vibration, noise sources, and electromagnetic interference areas. Try to avoid places with existing underground facilities.
5. Keep away from places that produce dust, fume, harmful gases, and produce or store corrosive, flammable, and explosive materials. The distance from the airport, landfill, river bank, shore or dam should not be less than 500m.
6. Keep at least 50m away from residential areas to avoid noise pollution.
7. Choose an open location according to the requirement of Figure 5-1 and Figure 5-2, and ensure that there are no obstacles from the surrounding area.
8. The minimum clearance between the racks should be back to back 100mm (0.5P System) or 500mm (1P System); 100mm(side by side) The clearances above are for reference only, the actual clearance should comply with the local standards and regulations and consider the maintenance feasibility.
9. For back-to-back arrangement, the minimum clearance between battery racks is 100mm, and the minimum clearance between the front side of the two racks is 2000mm, the clearance here refers to the road width, customer should consider the maintenance feasibility based on the site condition.
10. For back-to-enclosure arrangement, the minimum clearance between the back side of rack and the enclosure is 100mm(0.5P), 500mm(1P), and the minimum clearance between the front side of rack and the enclosure is 2000mm, the clearance here refers to the road width. (For installation and maintenance)
11. Considering the convenience of maintenance, CATL recommend the layout as below:

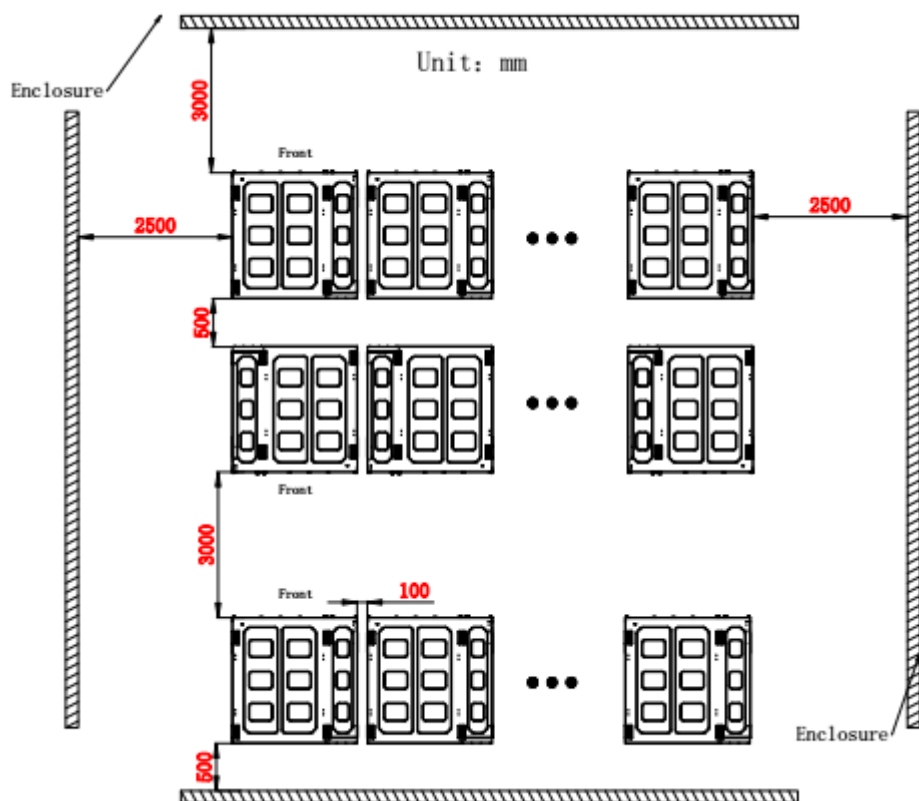


Figure 5-1: Clearance requirement for 0.5P&1P System

5.2 Foundation Requirement

Before installing this rack, build the foundation and trench on the selected ground. The requirements for foundation construction are as follows:

1. The size of foundation meets the requirements of rack installation and bearing capacity as following shown.

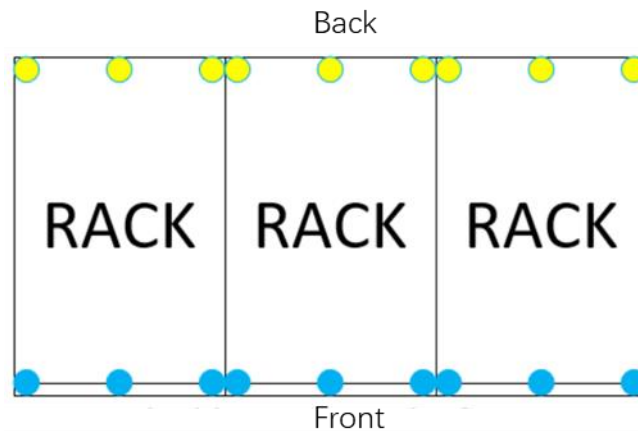
Table 0-1 Installation site requirement

Ground type	Condition requirements	Note
Concrete floor	The ground needs to be able to bear a load of 3.7 tons	Ground should fulfill: ◆ Level deviation $\leq \pm 10\text{mm}$ ◆ Flatness deviation $\leq \pm 4\text{mm}/2\text{m}$
Plain land surface		

2. Foundation requirement:
 - a. Fix the bottom foot of the rack,

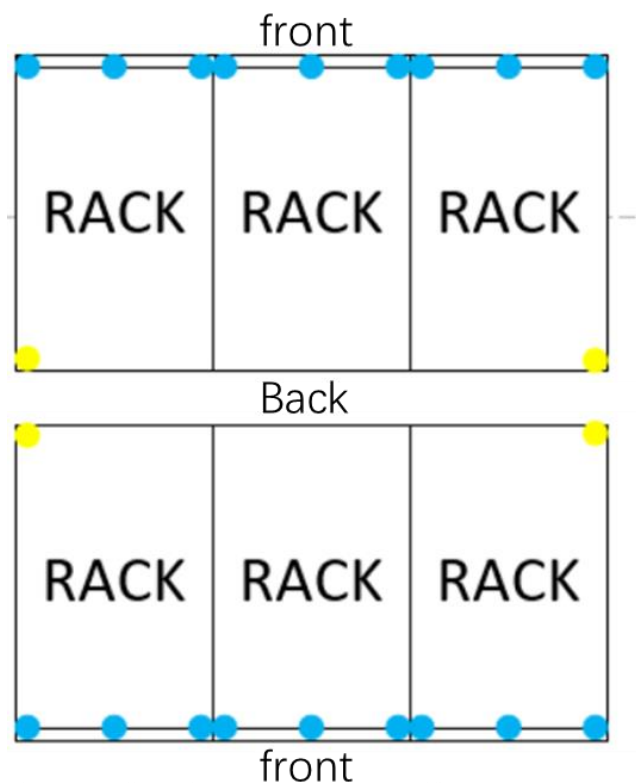
- b. The bottom of each rack is fixed with M16 Chemical anchor bolt. The torque of the torque wrenches is set to 150~200N.M.

Different layouts of racks are shown as Figure 5-1 and 5-2:



Note: The blue spots in the figure presents that point need to be locked with two bolts. The yellow spots in the figure presents that point need to be locked with one bolt.

Figure 5-3: Single row fixation



Note: The blue spots in the figure presents that point need to be locked with two bolts. The yellow spots in the figure presents that point need to be locked with one bolt.

Figure 5-4: Back to back fixation

3. Bury the ground grid and reserve a ground bar at the ground position of the rack. Connect one end of the ground grid to the embedded ground grid and the other end to the rack ground point. When the ground network is embedded, reserve enough length for the ground lug to connect to the ground point on the rack.
4. The grounding resistance of the rack is less than or equal to 0.1Ω .
5. The rack energy storage system uses underside cabling, so the cables need to be buried under the power cabin in advance.
6. The inner diameter of the protective pipe should not be less than 1.5 times the outer diameter (including the protective layer) of the cable.
7. Foundation construction should meet the local historical maximum rainfall drainage requirements. The discharged water shall be treated in accordance with local laws and regulations.

5.3 Preparation for installation

5.3.1 Unpacking

Unpacking steps are as follow:

1. Remove the packing belt and winding film on box.
2. Remove the front and back Packaging panel from the front side and back side respectively.
Then Remove the right side and left side panels.
3. Unscrew four nuts (M16*110) on the wooden pallet, remove the bolt and washer.
4. Move the rack out of the tray.
5. Unpacking completed.

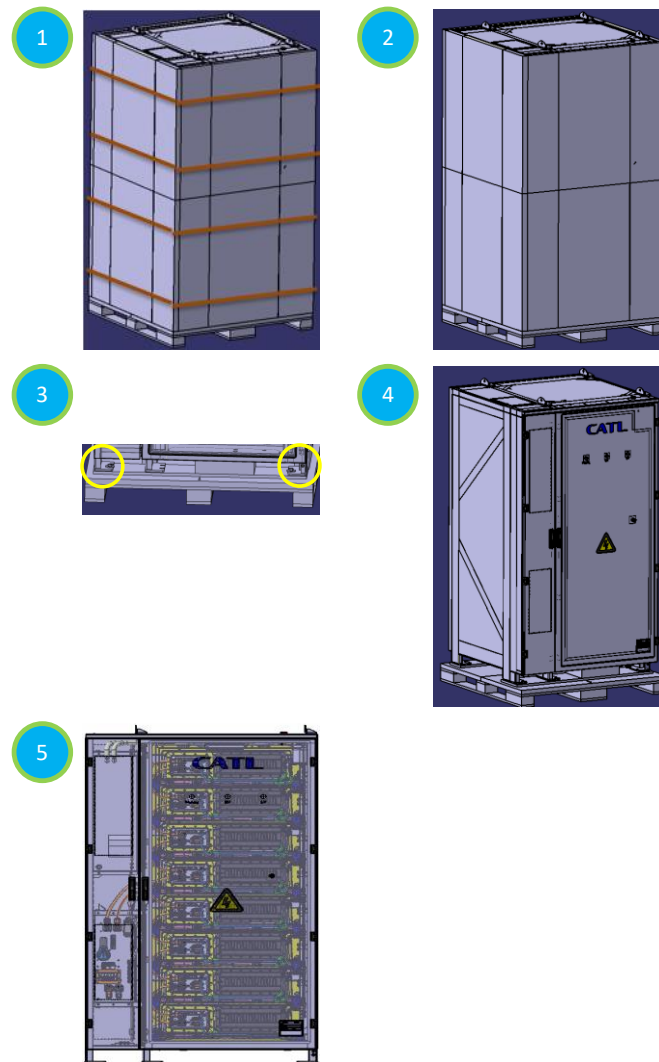
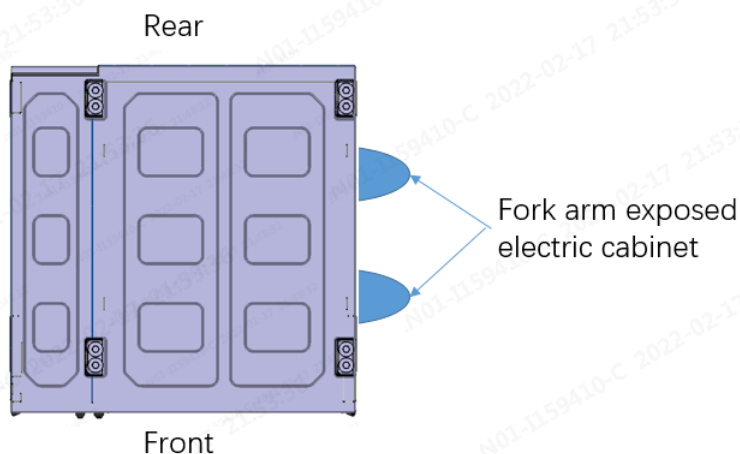


Figure 5-5: Flow chart of unpacking

5.3.2 Precautions for forklift transportation

1. The forklift arm needs to be protected to avoid dirt pollution on the fork arm, or the forklift scrapes the bottom of the cabinet.
2. Before the forklift fork arm reaches into the bottom of the cabinet, make sure the height of the fork arm is lower than the bottom of the cabinet to avoid collision with the cabinet.
3. After the forklift fork arm reaches into the bottom of the cabinet , make sure that the fork arm on the other side of the cabinet.
4. Forklift transport process should be maintained at a uniform speed.
5. Pay attention to the electric cabinet when transferring forklift truck.
6. Forklift model selection should consider the total weight of rack.



The forklift direction is for reference only

Figure 5-6: Forklift transfer reference

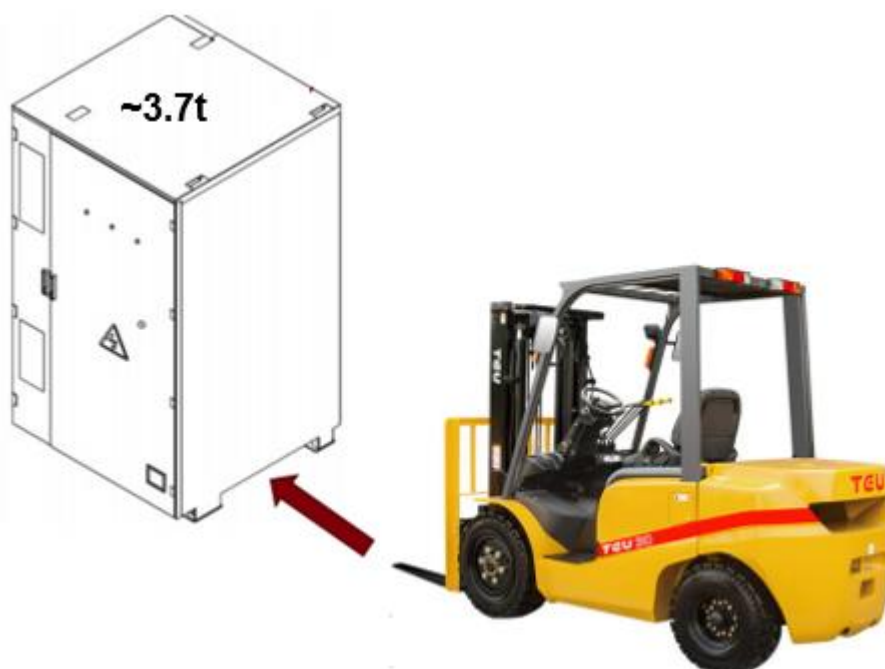


Figure 5-7: Forklift transfer recommend

5.4 Installation Procedure

Table 5-2: Installation procedure

Step	Procedure
1	Determine the position for installing racks based on the foundation design diagram;
2	Measure the support points of foundation racks fulfill the requirement of chapter 5.2
3	Use a crane to lift racks according to chapter 4.2, or use a forklift to transfer racks according chapter 5.3
4	Adjust the rack ensure that the support points at the bottom are evenly supported.
5	Secure the rack.

In the battery rack, the management system and a variety of sensing devices are installed. The whole battery system must be installed firmly and reliably without any loosening or shaking. An anti-collision device with sufficient strength must be installed on the periphery of the installation space of battery system so as to ensure that no safety accident (e.g. direct short circuit, overheating and combustion) will be caused by general collision which will not directly harm the battery system and batteries.

5.6.1 Electrical Block Diagram

Figure 5-8: Rack electrical block diagram

The battery rack has three main connection interfaces: DC power cable connection, AC auxiliary power connection and communication interface

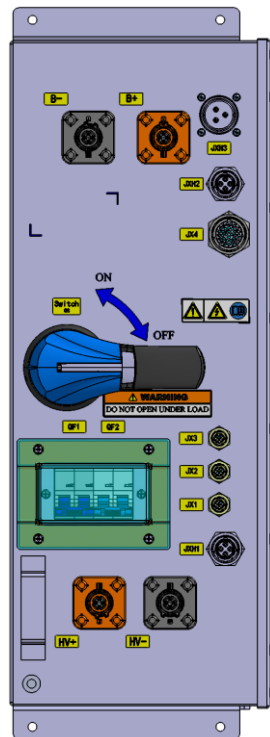


Figure 5-9: Front view of control box

5.6.2.1 Communication Cable Connection

1. Install the corresponding communication cable on JX2 and JX3:
2. In the first rack, JX2 is connected to the JX3 of the next rack, and JX3 is connected to MBMU.
3. In the middle rack, JX2 is connected to the JX3 of the next rack, and JX3 is connected to JX2 of the last rack.
4. In the terminal rack, JX3 is connected to JX2 of the last rack, and JX2 is connected to the terminal resistor connector.

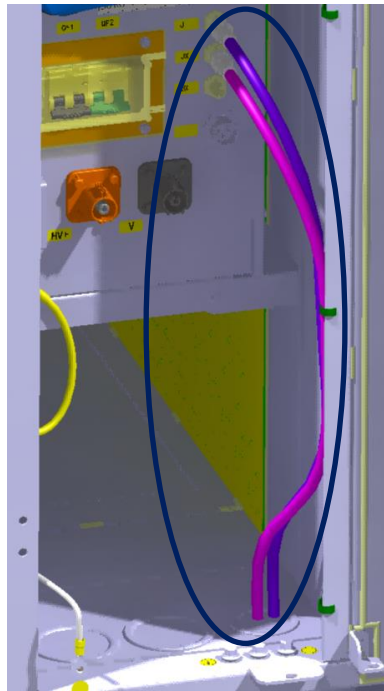


Figure 5-10: Communication Cable

5.6.2.2 Auxiliary Power Cable Connection

1. Install the auxiliary power cable (including the cable for chiller and BMS power supply) on JXH1.



Figure 5-11: Auxiliary power cable

5.6.2.3 DC Power Cable Connection

1. Isolation switch QS1 and air circuit breakers QF1, QF2 should be in "OFF" state.
2. Orange connector connects to HV+, black connector connects to HV-.
3. Push the plug of the HV connector into the receptacle during installation. When you can feel obvious reaction force, press the secondary lock(the disassembly method is opposite);

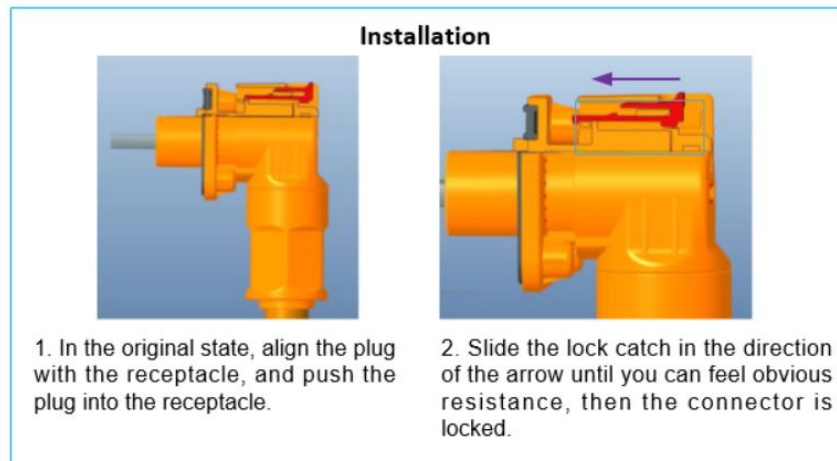


Figure 5-12: High Voltage Connector Installation

4. If the secondary lock cannot be pushed in, it indicates that the HV connector is not be installed correctly.

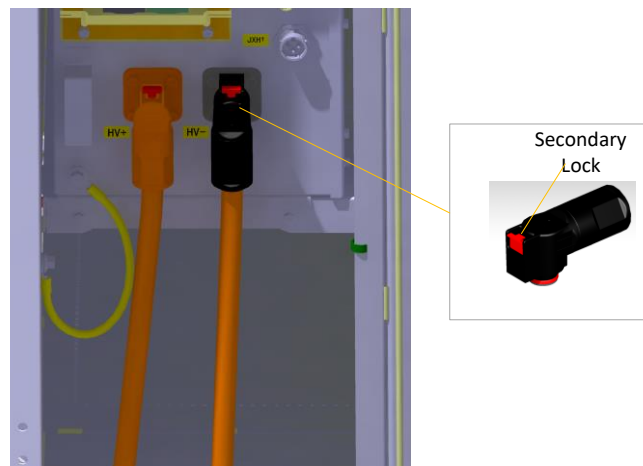


Figure 5-13: DC Power Cable

5.6.2.4 Grounding connection

Fix the grounding wire to the grounding point on the rack with M8 hexagon flange bolts. The torque for M8 fixation: 10N.M, and the size of the ground wire $\geq 50\text{mm}^2$ (cooper).

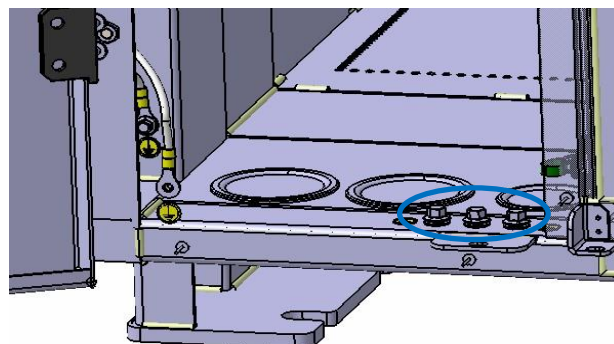


Figure 5-14: Grounding of the rack

5.6.3 Wiring Quality Requirements

1. Inspect whether the polarity of DC cables is correct, nuts are installed properly, and cable signs are correct.
2. Inspect whether the insulation impedance of all racks meet the requirements.
3. Inspect whether grounding conduction of grounding wires is good.
4. Before power on, check the connection cable of the whole system, and make sure that the cable connection is reliable without aging fracture and insulation damage.
5. Check whether all communication wires and cables at the connection end are tight and reliable;

5.7 MSD Installation

After confirming that the above installation is correct, install the MSD. During transportation process, the Manual Service Disconnect (MSD) from each battery module is always removed as the factory default status, so at the end user site, the MSD must be installed on each battery module. The steps of MSD installation are as following:

1. Make sure that isolation switch QS1 and circuit breakers QF1, QF2 should be in "OFF" state.
2. Open the battery compartment door.
3. Install 8 MSDs upper cover on the panel of each module box.
4. Put the handle in the vertical state and align the cover of the maintenance switch with the receptacle.
5. Push it in and rotate the handle.
6. When you can hear the "da da" sound, push in the secondary lock(the disassembly method is opposite), and the installation is completed.
7. Using multimeter measure the B+, B- interface of the battery module, if voltage range between 145.6-187.2Vdc, it is lockable; if voltage is 0Vdc, it is unlockable

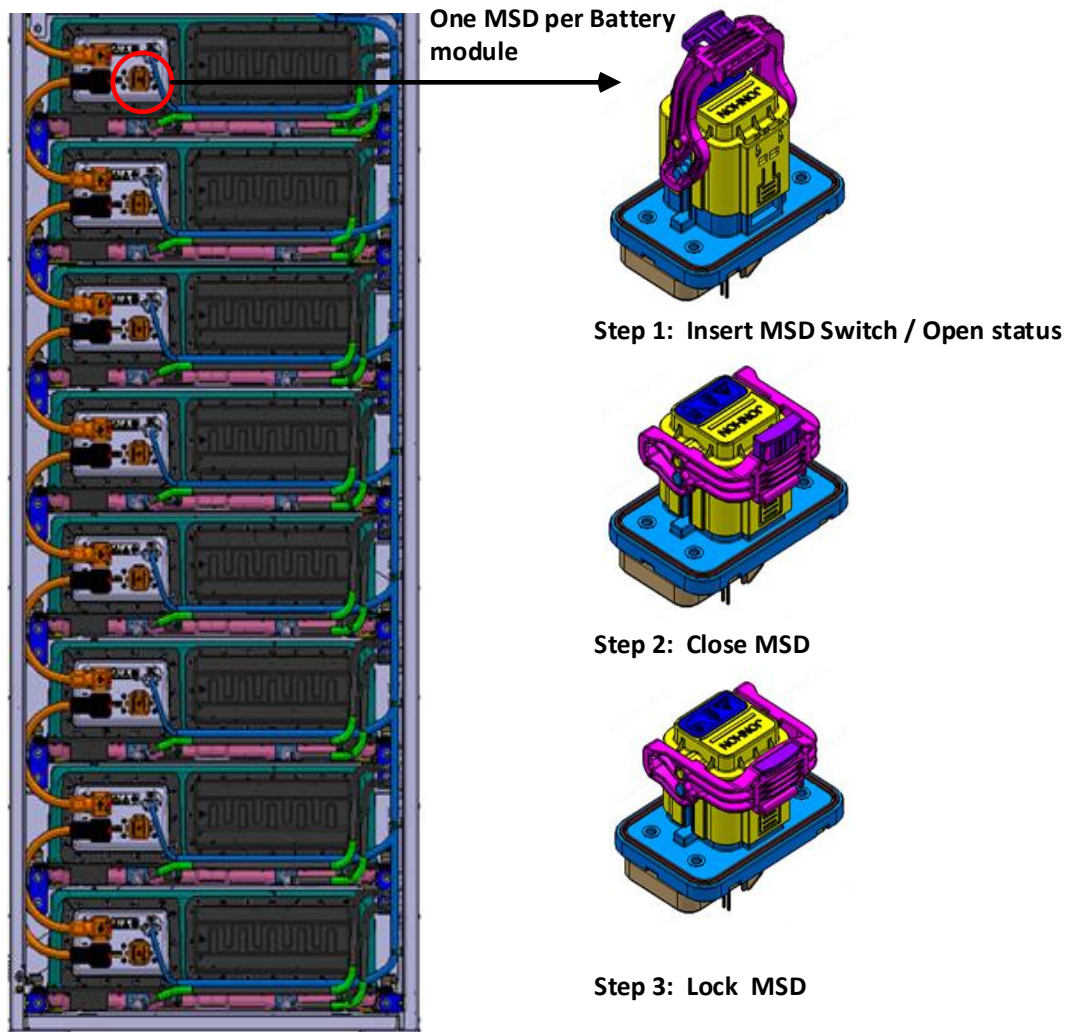


Figure 5-15: MSD installation

5.8 Wiring Attention

1. JX1 is a debugging interface, which needs to be kept in a sealed state.
2. The connector need to be tighten during the installation of JX2 and JX3.
3. To ensure the high voltage connector is locked.
4. To ensure the MSD is locked.
5. Strictly prevent any form of short circuit during the connection process.
6. It is strictly forbidden for the operator to operate without training.
7. It is strictly forbidden for the operator to operate freehand without wearing protective equipment.
8. All connections must be made under explicit guidance, and any form of guessing and vague trial operation methods are strictly prohibited.
9. The key points of the connection are: ensure the connection is correct, reliable (will not be loosened), good contact, no short circuit.
10. After the connection is completed, it must be measured and confirmed point by point.
11. Before wiring, make sure QF1, QF2 and QS1 in "OFF" state.

5.9 Cable Connection Confirmation

Before power on, check the connection cable of the whole system, and make sure that the cable connection is reliable without aging fracture and insulation damage.

5.9.1 Cable Checking

1. Visual inspection
2. Use multimeter buzzer measure both ends of the cable, if cable is ok, the buzzer will make a sound; if cable is breake, there is no sound.
3. Check whether the positive and negative poles of the DC output power cable in the battery rack are connected correctly;
4. Check whether the power connection of the battery cupboard is correct;
5. Check whether all communication wires and cables and sub connections at the connection end are tight and reliable;

5.9.2 Status Checking of Equipment Switch

Close circuit breaker QF1, QF2 and isolation breaker QS1 in “OFF” state (red frame in above figure).

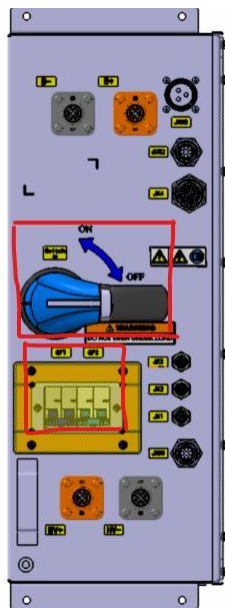


Figure 5-16: Front view of control box

6 CATL Monitor Software

As the LFP battery energy storage running monitor, CATL Monitor Software can realize following fundamental functions, and the installation process and instruction can refer to follow chapters

1. Data monitoring and recording
2. Rack numbering
3. CSC numbering
4. Rack on-off controlling
5. Fault Alarm checking and analyzing
6. Manual Balancing

6.1 Commissioning Tool

For CATL Monitor Software installation, following commissioning tools need be installed.

Table 6-1: Commissioning tool

Tool	Description	Overview
CAN Tool	CAN Tool provides connections between CATL BESS and the Monitor software in host computer. Flashing and configuring are both done by these tools. CATL Order Number: 380240-00002	
Commissioning cable	Commissioning cable provide the cable connector to control box and CAN Tool CATL Order Number: 550250-17957	

6.2 Commissioning system hardware setup

As the debug interface, JX1 connector in EnerOne Control box can be used for external commissioning. After CAN Tool Driver installation, the CAN bus of control box can be accessed by PC.

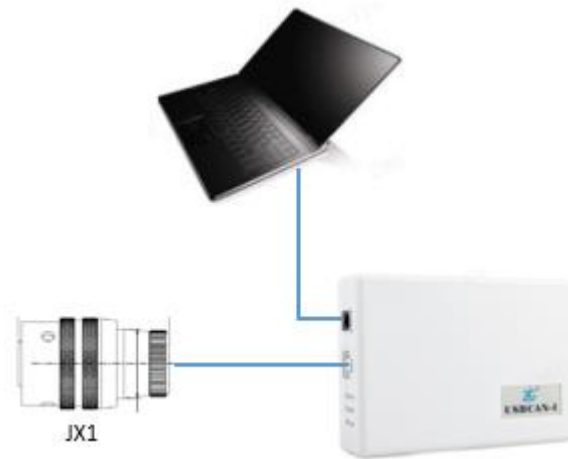


Figure 6-1: Commissioning system setup

6.3 Software configuration

Switching on the auxiliary power supply firstly, if the green indicator of door lights up, it means the 24V power supply of SBMU is ok. Then use CAN tool software to verify the MCAN log. After getting normal log, start verification and flashing MBMU and ETH software, and then configuring ETH IP according to following steps:

6.3.1 MBMU and ETH status verification

- a) Open ETH IP setting tool, select CAN tool type, press open CAN type, set and read IP address, and then it should show 'successful' after completion. If failed, please check CAN loop and retry. After 'successful' notification appears, Ping IP address and check the connection.

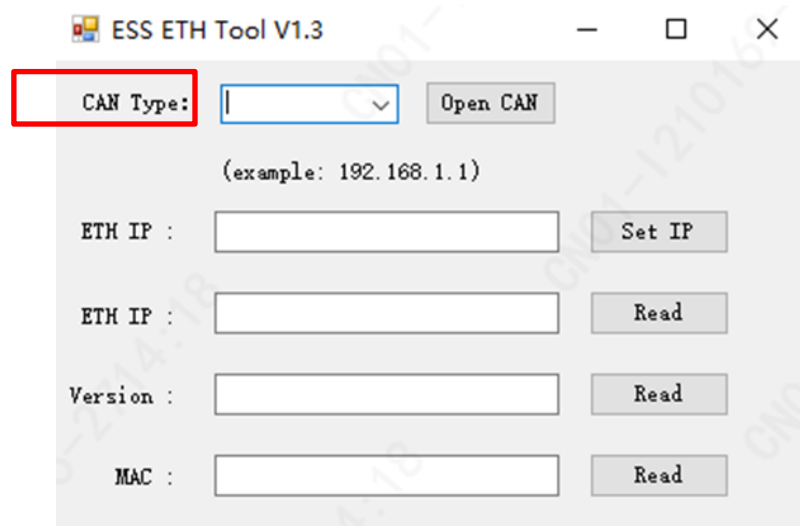


Figure 6-2: ETH IP setting tool interface

If not able to set IP:

1. Use CANtest to read the log

2. Check ETH power supply and CAN_H/CAN_L loop connection.
 3. Only turn on the ETH power supply and configure IP.
- b) After verifying the version, configure IP using setting tools (if the version is wrong, flash the ETH software first.)

6.3.2 MBMU software flashing

Step	Description
1	Open MBMU/ETH flash tool
2	Select 'setting', 'ZLGCANII', and click 'save'
3	Select driver path in source file configuration <u>\\CATL BMS Flash Tool 1.25 For ESS MBMU 201208\\Configuration\\ESS_MBMU_201028\\FlashDriver</u>
4	Select 'APP path', and select MBMU hex file (configured project by project) Such as: <u>AMBMUF067Q00CT03_SA_200501.hex</u>
5	Select 'system', click 'run'.
6	Click 'flash' and wait until 100% done.

6.3.3 ETH software flashing

Step	Description
1	Open MBMU/ETH flash tool
2	Select 'setting', 'ZLGCANII', and click 'save'
3	Select driver path in source file configuration <u>\\CATL BMS Flash Tool 1.25 For ESS MBMU 201208\\Configuration\\ESS_MBMU_201028\\FlashDriver</u>
4	Select 'APP path', and select MBMU hex file (configured project by project) Such as: <u>AMBUMUF067Q00CT03_SA_200501.hex</u>
5	Select 'system', click 'run'.
6	Click 'flash' and wait until 100% done.

6.4 CATL Monitor System

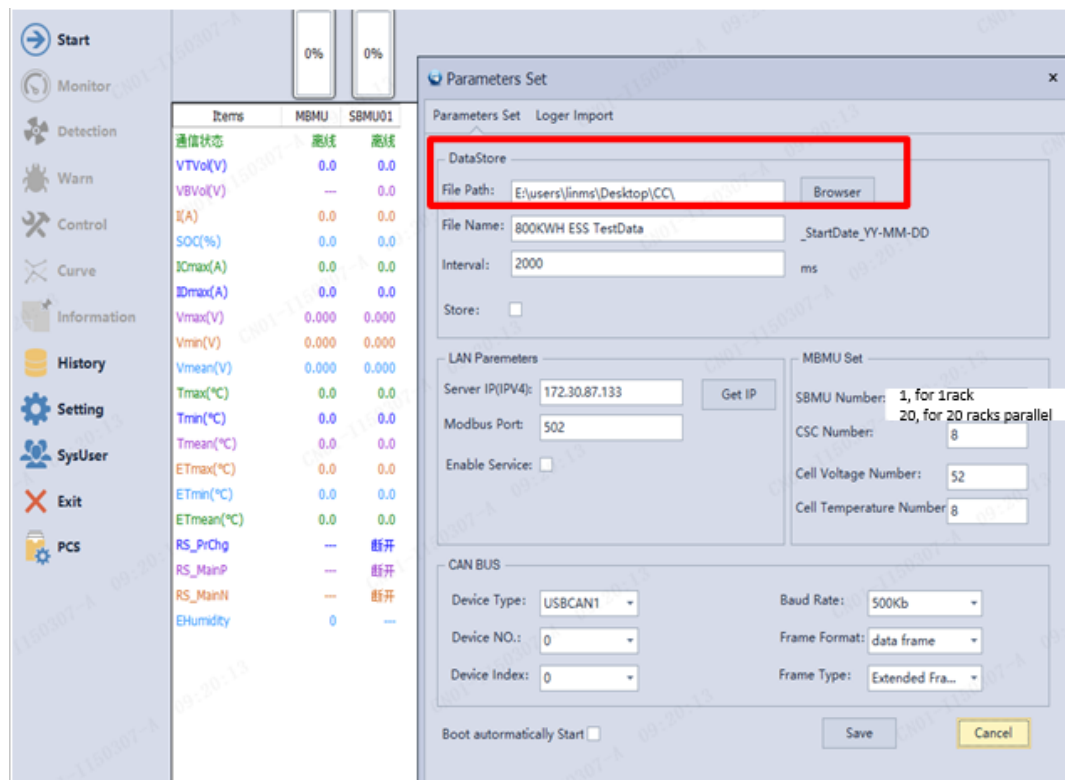
Table 6-2: CATL Real-time Monitoring System Overview

Step	Description
1	CATL Real-time Monitoring System installation After connecting CAN tools to the computer, open the real-time monitoring system

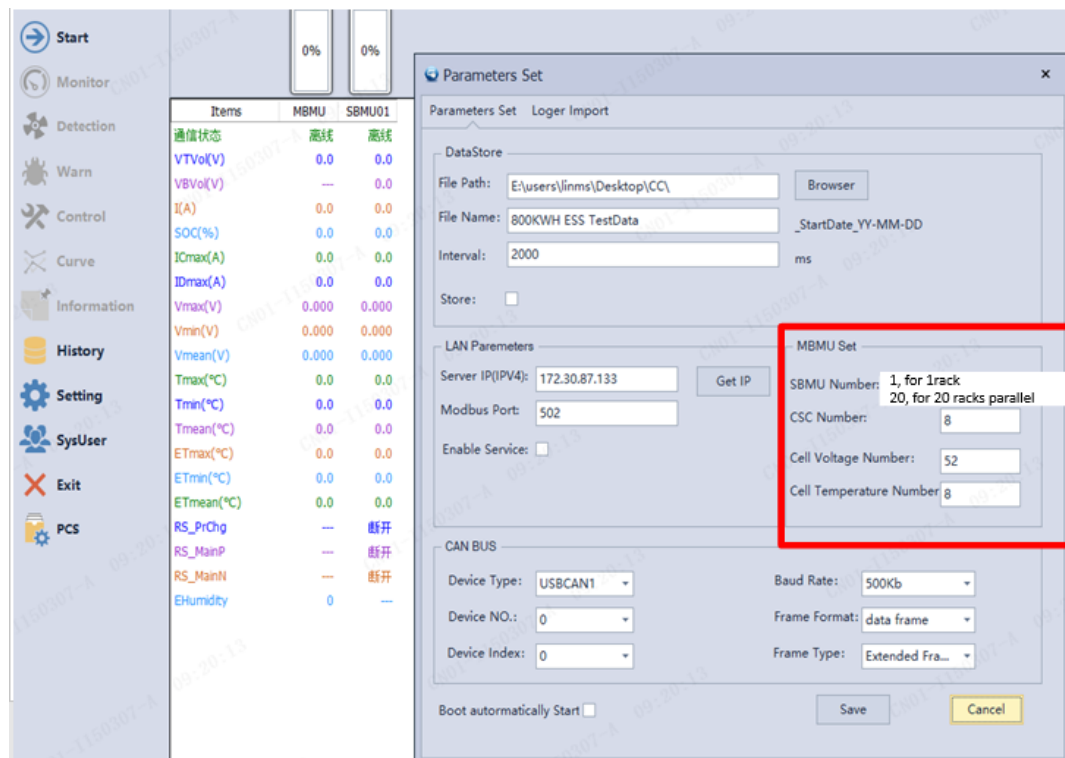
2

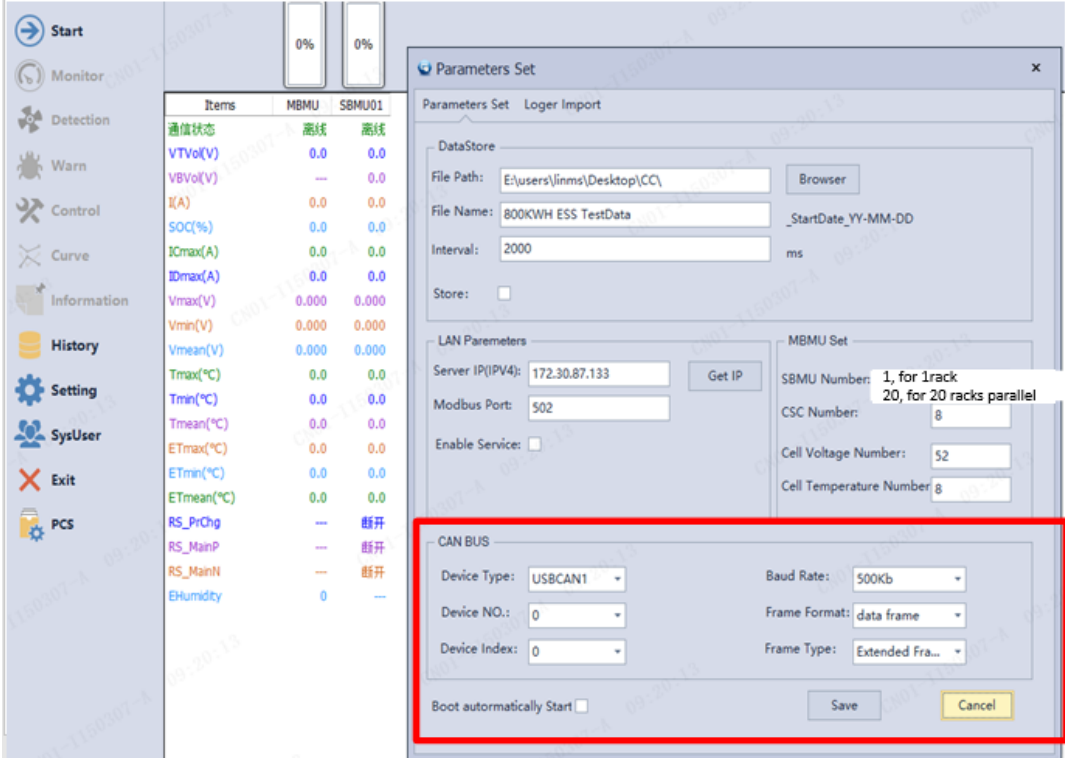
Data Store file path setting

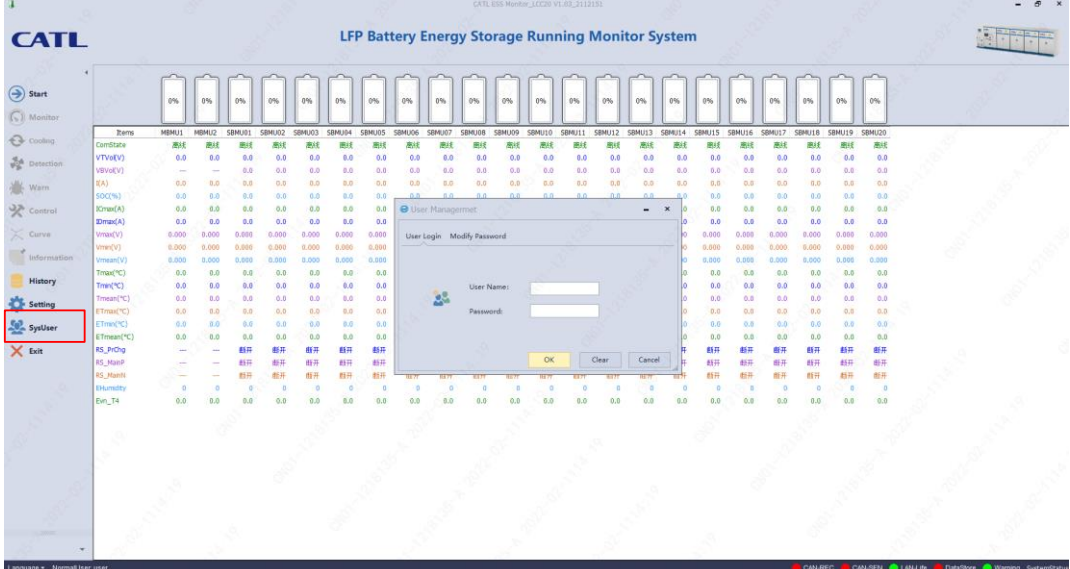
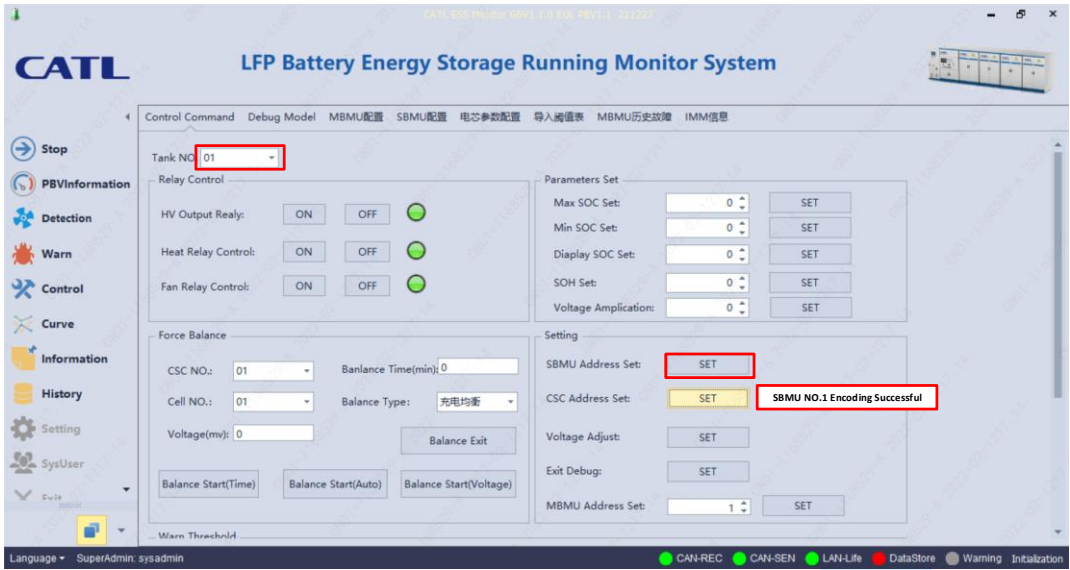
- Set up the real-time monitoring system by choosing the data storage place, after that, all shown data on the monitoring system will be stored on the path.



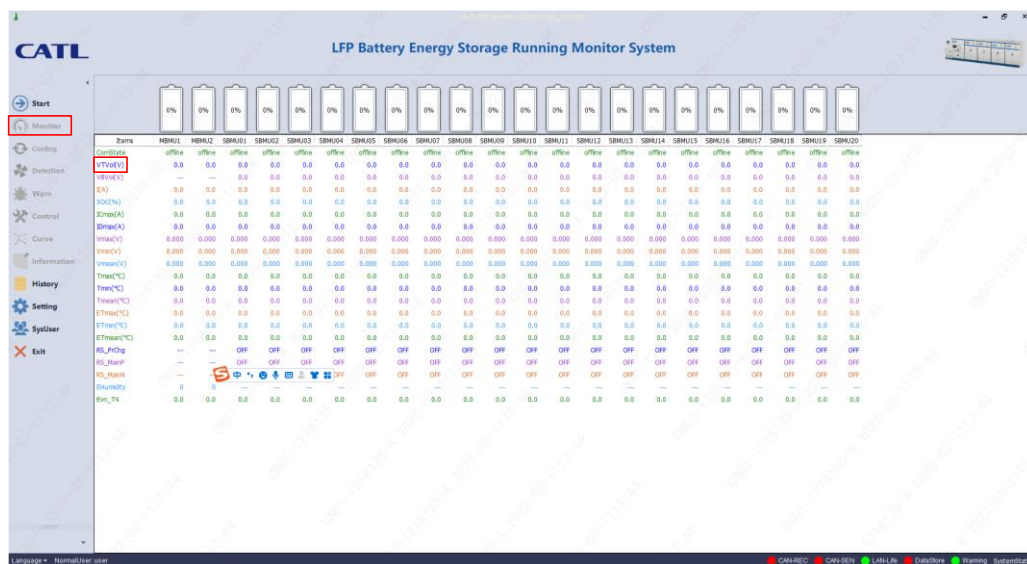
3

MBMU Setting:

	● SBMU number: the number of racks on the same CAN circuit, for single rack, SBMU number is 1, for 20 racks parallel connection, the number must be updated to 20; ● CSC number: 8, the number of modules in a single rack; ● Cell voltage number: 52, the number of cell in a single module; ● Cell temperature number: 8, the number of temperature test points in a single module;
4	Interface setting: ● Regarding device type, usually, choose USBCAN1 if using ZLGCAN (ZLGCAN has two interface, CAN1 and CAN2). If using ValueCAN tool, then choose ValueCAN. ● Regarding Baud Rate and data type, leave them as the default value. Click 'save', and the monitoring system will restart automatically. If others Baud Rate setting is required, the definition is according to the type of CAN, the value of ACAN is 250k, CCAN, MCAN, SCAN need chose 500k 
5	User login interface setting ● Click the menu SysUser, the user login interface will pop up. ● The User Name is sysadmin, the password is 1234567890111. Then click the OK.

	
6	SBMU&CSC Encoding - When encoding the first SBMU, set the Tank NO. to "1", and then click the "SBMU Address Set", after that, click "CSC address set". If encoding successful, following message "SBMU No.1 encoding successful" will be got. - Others SBMU&CSC encoding follow same process 
7	Rack on-off controlling After confirming that the condition for closing High Voltage relays is fulfilled, click 'ON'. Then all the racks will be powered on.

8	Internal voltage calibration (have done in factory setting, just need for maintenance) - After closing high voltage relay, BMS will measure system voltage and accumulative voltage of single cell. For limiting these two voltage difference smaller than 2V, the internal voltage calibration need be done. - Firstly, choose the Tank No. correctly, and then click the 'set' in voltage adjust.
9	Operating status Monitor Select "monitor", check if all the batteries are on closed state, and check VTVOL to see if there is voltage data



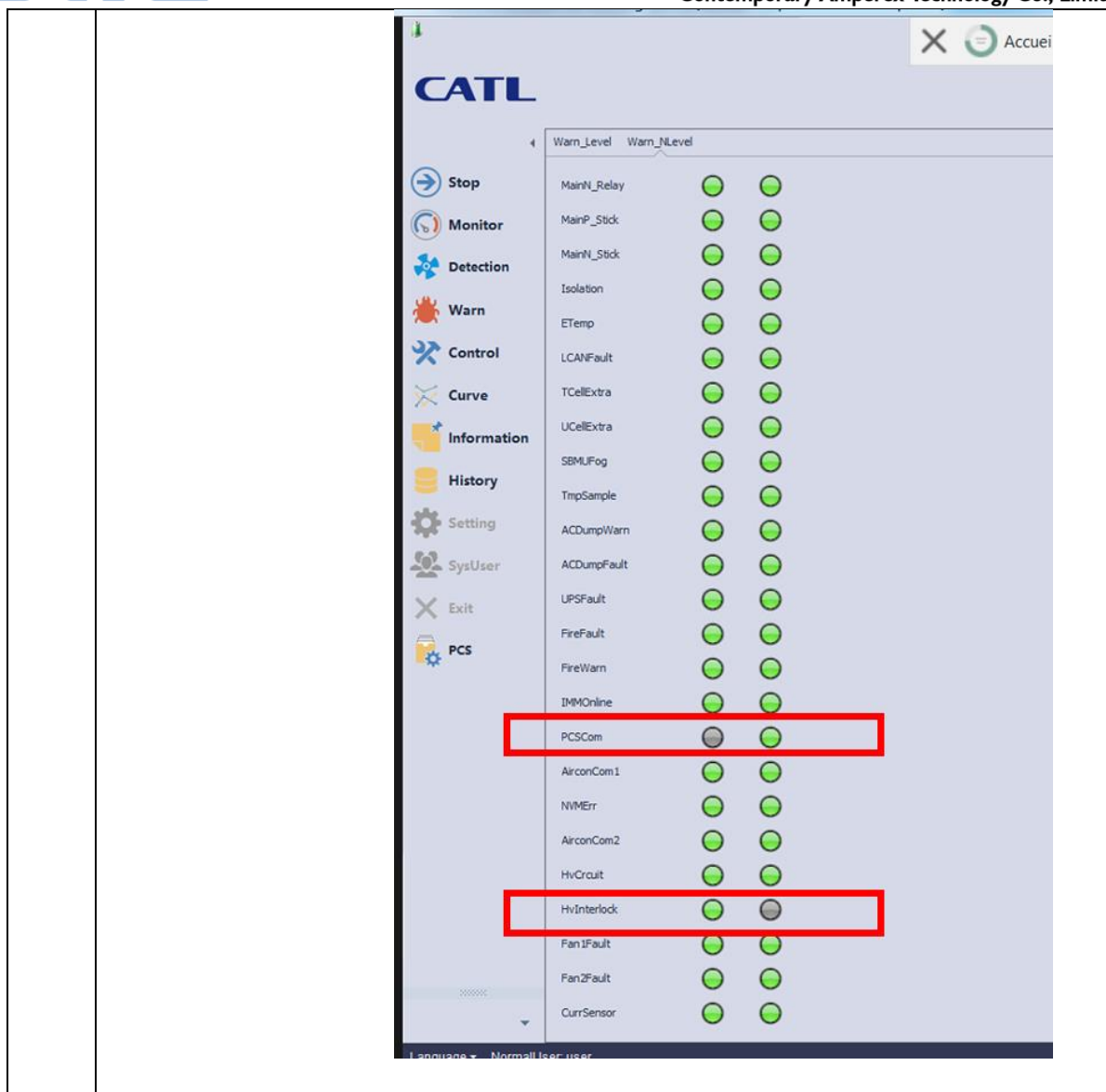
- Check if allowable charging/discharging current appears. If so, it means that the system can charge and discharge normally.



10

Warning status Monitor

- Select 'warn'. Alarm status can be found here. Based on the alarm status, figure out the corresponding problems and do the analysis.



7 Commissioning

7.1 System inspection before commissioning

In accordance with the requirements in the system instructions, patrol inspection should be performed for the system and execution and recording should be carried out as per the following table.

Table 7-1: Checklist of system inspection

No	Test item	Description	Result	
1	Appearance Inspection	The appearance of product has no damage and deformation; paint coating has no raise and damage	OK ☐	NG ☐
			OK ☐	NG ☐
			OK ☐	NG ☐
2	Check bolts	Whether bolts get damaged or loosed	OK ☐	NG ☐
3	Check HV disconnection	Use the multi-meter to test open voltage = 0V;	OK ☐	NG ☐
		Use the multi-meter to test the resistance between positive and negative poles is = ∞ .	OK ☐	NG ☐
4	Check LV cable connection	Whether cable connection is correct according to chapter 5.8	OK ☐	NG ☐

	WARNING
	In the battery system, all power connections must ensure that a sufficient number of protective measures for insulation will be taken, the positive and negative poles of a battery will in no way contact the outer box to lead to leakage of electricity and a short circuit, and the positive and negative poles of the battery system will not be short-circuited in any case; otherwise, a major safety accident and an electric shock might be caused.

7.2 Commissioning flow chart

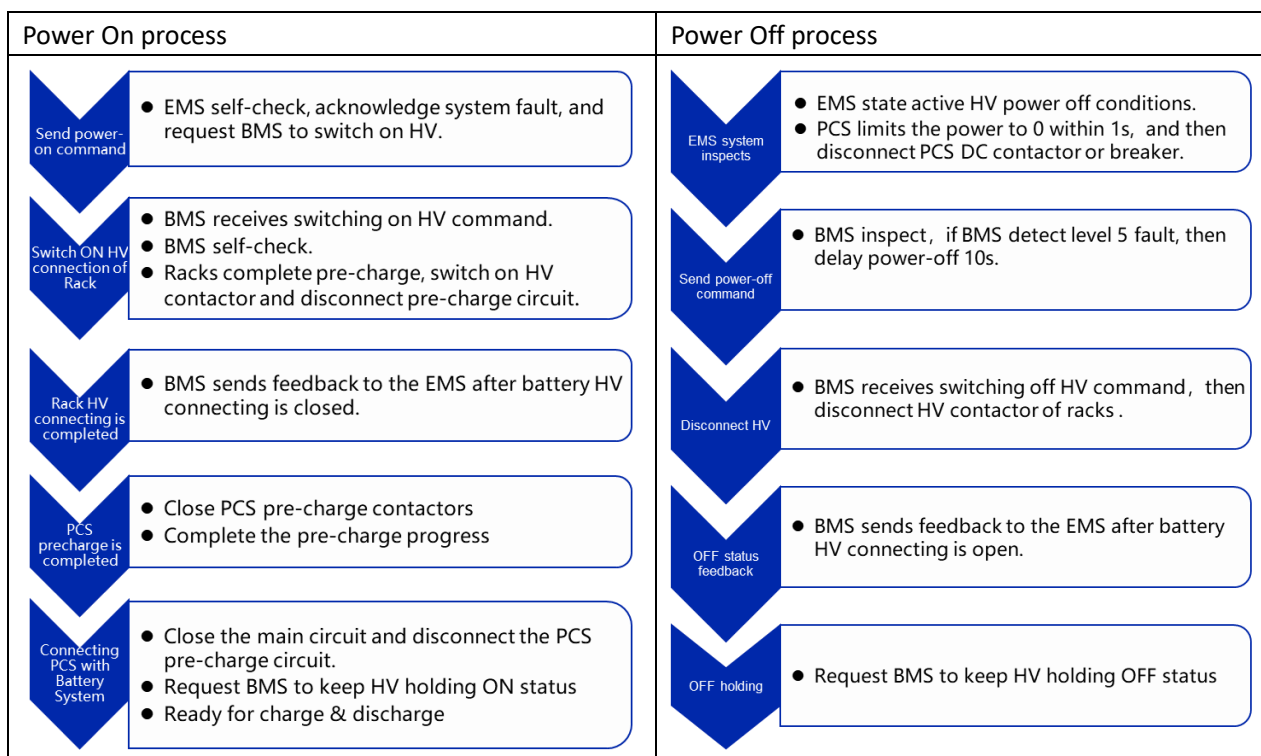
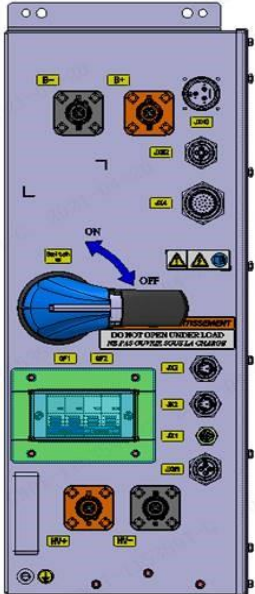


Figure 7-1: Commissioning flow chart

7.3 Switch on HV relays of battery control box

7.3.1 Switch on insulation switches of control box.

Table 7-2: Control box overview

Control box	Operating & Status
	Switch on insulation switches QS1

7.3.2 Switch on HV relays

After the auxiliary power supply is switching on, use the close HV relays command to close the HV relays according to communication protocol. At this moment, the positive and negative output ends of the battery racks will generate about 1164V~1497.6Vdc high voltage. The light combination will correspond to the following state:

Table 7-3: Indicator light: High voltage power supply

System state	Indicator light for the low voltage circuit ●	Indicator Light for the high voltage circuit ●	Alarm indicator light ●
Power OFF	Yellow light is OFF	Red light is OFF	Yellow light is OFF
Failure	Green light is ON	Red light is OFF	Yellow light is ON
power on of low voltage circuit	Green light is ON	Red light is OFF	Yellow light is OFF
Power on of high voltage Circuit	Green light is ON	Red light is ON	Yellow light is OFF
Charge/Discharge	Green light is ON	Red light is ON	Yellow light is OFF

If the combination of the lights on and off is inconsistent with the above list, it means that the system is not properly powered on. Repeat the steps;

If the system fails to start up after repeating the steps for 5 times, please conduct preliminary troubleshooting of system failure according to the “**Table 8-2: List of Failure diagnosis procedures**”;

If the problem cannot be solved, please contact the manufacturer for system troubleshooting.

7.4 Startup PCS and Precharge DC capacitors

Due to PCS include DC capacitors for filter function, the precharging process is always mandatory, it's recommend that PCS precharging is realized by PCS instead of DC battery system.

7.5 Switch on the connection between PCS and Rack

For EnerOne rack, high current will be conducted to PCS during charge/discharge process, for line protection and controlling function, a DC breaker with motorized controlling need be used for switching on/off the connection between PCS and EnerOne rack.

After startup PCS and precharging of DC capacitors, then switch on DC connection.

7.6 EMS send command to PCS to start charge/discharge

7.6.1 Charging of Battery System

When the connection is closed between PCS and EnerOne rack, enable PCS (EMS) working in **constant power control model** and start charging, the BMS detects the charging current, and the battery system is charged, at this time, the combination of the on and off of each indicator will correspond to the state as Table 7-3.

7.6.2 Discharging of Battery System

When the battery system is charged at high voltage, if the battery system detects the discharging current, the battery system will enter the discharging state and the battery system will start discharging. At this moment, the combination of lights on and off will correspond to the state as Table 7-3.

8 Diagnosis of common abnormal problems

8.1 System status word

A UNIT16 status word had been defined in Modbus Map, the system status can be read by BMS via register address.

8.2 Countermeasures of system alarm

If system failure mode is detected, the abnormal status will be divided into 5 level to distinguish the severity of failure, and then the system will automatically switching to a safety operating model according to following definition.

Table 8-1: Treatment measures of failure at different levels

Failure level	Severity degree	Alarm to EMS	BMS failure dry contact	HV Contactor response
CAT 1	Very Low	Report	close	No action
CAT 2	Low	Report	close	No action
CAT 3	Medium	Report	close	No action
CAT 4	High	Report	close	Outage of failure single rack
CAT 5	Very High	Report	open	Outage of all battery racks

8.3 Countermeasures of abnormal conditions

When enable the battery rack, if it cannot work normally, please check the potential cause according to **Table 8-2**. At the same time, pay attention to check whether it is caused by external environment, such as temperature, noncompliant humidity or load overload. If the maintenance or replacement of spare parts is involved, the after-sales service procedure should be followed. Meanwhile, the disassembly and installation are not allowed without authorization.

This chapter contains only a few simple failure diagnosis procedures. If the answer to the diagnosis is not very unequivocal, or the information obtained is not enough to settle the problem, please contact the local office or distributor of the CATL.

Table 8-2: List of Failure diagnosis procedures

S/N	Failure or alarm phenomena	Possible cause of failure	Troubleshooting scheme
1	The green indicator light of the battery rack is not on	1. Poor contact of wiring harness	Check indicator light wiring harness
		2. SBMU failure	Replacement of SBMU
		3. Breakdown of indicator	Replacement of indicator light
2	The yellow indicator light of the battery rack is not on	1. Poor contact of wiring harness	Check indicator light wiring harness
		2. SBMU failure	Replacement of SBMU
		3. Breakdown of indicator	Replacement of indicator light
3	The red indicator light of the battery rack is not on	1. Poor contact of wiring harness	Check indicator light wiring harness
		2. SBMU failure	Replacement of SBMU
		3. Breakdown of indicator	Replacement of indicator light
4	Pre-charge contactor cannot be closed	1. Poor connection of relay wiring harness	Check wiring harness connection
		2. Contactor failure	Replacement of pre-charge contactor
		3. SBMU failure	Replacement of SBMU
5	Main positive DC-contactor cannot be closed	1. Poor connection of main positive DC-contactor wiring harness	Check wiring harness connection
		2. Main DC-contactor failure	Replacement of main DC-contactor
		3. SBMU failure	Replacement of SBMU
6	Main negative DC-contactor cannot be closed	1. Poor connection of main negative DC-contactor wiring harness	Check wiring harness connection
		2. Main negative DC-contactor failure	Replacement of main negative DC-contactor
		3. SBMU failure	Replacement of SBMU
7	Adhesion of main positive DC-contactor	1. Damage of main positive DC-contactor	Replacement of main positive DC-contactor
8	Adhesion of main negative DC-contactor	1. Damage of main negative DC-contactor	Replacement of main negative DC-contactor
9	Abnormal power down of CSC	1. Abnormal CSC power supply wiring harness	Check CSC power supply wiring harness
		2. CSC failure	Replacement of CSC

S/N	Failure or alarm phenomena	Possible cause of failure	Troubleshooting scheme
10	Abnormal cell voltage detection	1. Poor connection of voltage detection wiring harness	Check wiring harness connection
		2. CSC failure	Replacement of CSC
11	Abnormal NTC temperature detection	1. Poor connection of temperature detection wiring harness	Check wiring harness connection
		2. Temperature sensor failure	Replacement of temperature sensor
		3. CSC failure	Replacement of CSC
12	Abnormal total voltage detection	1. Poor connection of total voltage detection wiring harness	Check wiring harness connection
		2. SBMU failure	Replacement of SBMU
13	Internal communication anomaly	1. Poor connection of internal CAN wires	Check CAN connecting wires by pluck and insertion
14	Anomaly caused by unconfigured system parameters	1. Unconfigured battery system parameters	Configure system parameters
15	Current sensor anomaly	1. Poor connection of 12V power harness wiring	Check wiring harness
		2. Poor connection of SCAN communication wires	Check wiring harness
		3. SBMU power generation anomaly	Replacement of SBMU
16	SBMU automatic shutdown	1. Poor connection of MBMU wake-up signal wires	Check wiring harness
		2. MBMU failure	Replacement of MBMU
17	Cell over-voltage alarm	1. Poor equalization	Stop charging and standby equalization
		2. System overcharge	Stop charging
18	Cell under-voltage alarm	1. Poor equalization	Stop discharge and standby equalization
		2. System over-discharge	Stop discharge
19	NTC over-temperature alarm	Excessive temperature of individual NTC	Check whether the air conditioner is in operation
		Excessive charging-discharge current	Appropriately reduce charging and discharge current
20		1. Lower temperature of	Check whether the

S/N	Failure or alarm phenomena	Possible cause of failure	Troubleshooting scheme
	NTC under-temperature alarm	individual cores	heating function is on (if the equipment cabinet has heating function)
		2. Lower ambient temperature	No charging
21	Charging over-current alarm	1. Excessive current of system charging	Reduce charging power
22	Discharge over-current alarm	1.Excessive current of system discharge	Reduce system load
23	Alarm of higher SOC	1.The system power is full (overcharged)	Stop system charging
24	Alarm of lower SOC	1.The system power is too low (over-discharge)	Stop system discharge
25	Alarm of higher monomer temperature difference	1.Non-uniform heat dissipation of equipment cabinet	Check whether the air conditioner is on
		2. Temperature sensor failure	Replacement of temperature sensor
26	Alarm of higher monomer voltage difference	1.Excessive difference of cell capacity	Battery system can be charged to over 80% SOC, with standby equalization
27	SOC difference alarm	1. The electric quantity difference of different battery racks is too large	— —
28	Abnormal alarm of equalization circuit	1. Equalization abnormal CSC board	Replacement of CSC board

8.4 After-sales Service

CATL will provide clients with a full range of technical support and after-sales service. Users can gain services by dialing our service number. Service Number: **86 400-918-0889**

Please refer to the contract for the free warranty service information.

The following circumstances are not within the scope of our free warranty service:

1. System damage or failure caused by not following the user manual.
2. Damage or failure caused by not following the relevant electrical safety specifications for wiring and power supply or caused by poor site environment.
3. System damage or failure caused by users' private modification.
4. System damage or failure caused by irresistible natural factors, such as typhoon, earthquake, flood, fire or harsh environment (high temperature, low temperature, high humidity, acid rain,

etc.).

5. After the failure occurs, the user fails to maintain the initial failure state, fails to timely notify the manufacturer and handles without authorization, thus causing it is unable to make a practical fault identification of the failure causes.

9 Product Maintenance

During whole product lifecycle, abnormal operating must be avoid, regular maintenance in Table 9-1 is mandatory to ensure safe and reliable operation and acquire an optimal performance for system.

Table 9-1: Product maintenance category

Component Category	Service Descriptions	Frequency
Thermal Management System	Thermal management system inspection	Quarterly
	Thermal management system maintenance	Annual
	Clean air filters	Quarterly
Fire Safety System	Fire alarm and detection system inspection	Annual
	Fire alarm and detection system maintenance	Annual
	Fire suppression system Inspection	Semi-Annual
Battery system	Battery maintenance (system check, cell balancing)	Annual
	Battery cable, appearance, grounding, dust removal	Annual
	Inspect battery management system alarms	Annual
	Visual inspection of all electrical terminations using thermal imager	Semi-Annual
Enclosure	Rust, bent structure	Annual
	Inspect cable entry, grounding, sealing, dust removal	Annual
	Inspect rack Painting, label, indicator	Quarterly
Reporting and Diagnostics	Inspection, maintenance reports and recommendations	Monthly
	Technical support and fault diagnosis	As required
	Data server connectivity check	Monthly
	Inspect battery running status, provide operating recommendations and maintain battery	Monthly

9.1 Battery System Maintenance

■ 9.1.1 Maintenance instructions before substation competition

We recommend checking the battery voltage at least every 6 months and if the battery voltage (minimum

cell voltage in the rack/module) lower than 3.186V, the battery rack/module needs to be re-charged for further storage. It is recommended to the customers to charge the battery rack/module with 0.25P to certain voltage level as below:

- Either a) charge to cell voltage $\geq 3.355\text{V}$,
or b) cell voltage $\geq 3.294\text{V}$ after 2 hours resting.

9.1.1 Maintenance instructions for normal operating system

Perform battery maintenance according to following plan on the system every twelve months to prevent battery damage.

1) Plan 1 (This plan is applicable when SOC of the battery system is low)

- Discharge the battery system to the cut-off condition (the lowest voltage $< 2.8\text{V}$), then stop discharging, standing for 1 hour.
- Full charging automatically to the battery system (The highest voltage $> 3.6\text{V}$), after charging, standing for 1 hour.
- Discharge the battery system to 30% and stop

2) Plan 2 (This plan is applicable when SOC of the battery system is high)

- Full charging automatically to the battery system (The highest voltage $> 3.6\text{V}$), after charging, standing for 1 hour.
- Discharge the battery system to the cut-off condition (the lowest voltage $< 2.8\text{V}$), then stop discharging, standing for 1 hour.
- Charge the battery system to 30% and stop

	CAUTION
	1. Check to ensure environmental safety, system safety, no alarm, no fault before performing maintenance operations. 2. After the battery maintenance of ESS is completed, it's suggested to notify CATL after-sales engineer to perform data analysis for free.

9.1.2 Maintenance instructions for short duration idle mode after COD (This plan is applicable for short time storage 0-6 months)

- The SOC should be kept at least above 5% (cell voltage $\geq 3.186\text{V}$) when idle.
- Customers should notice that the monthly battery self-discharge rate is $\leq 3\%$ and the function module in BMS will draw $\leq 9\%$ SOC from the battery side when idle. If the battery rack needs to be staying in standby mode after COD, it needs to be re-charged to certain level which can cover the 3% self-discharge rate and 9% BMS power consumption to avoid over-discharge.

	CAUTION
	1. The minimum cell voltage should not be less than 2.5 V in any case; otherwise, the warranty will be invalid. 2. The minimum cell voltage should not be less than 2.0 V in any case; otherwise, the battery will be over-discharged and may not be able to be repaired.

9.1.2 Maintenance instructions for long duration idle mode after COD date

- Suggested SOC range of battery storage: 30%~60%.
- If the battery system keep on idle mode more than 180 days, the battery is suggested to have SOC equal or above 50%. 3.186Vdc shall in any case be considered as a minimum voltage level for per cell, in case that this voltage level is reached, battery system must be recharged immediately. It is recommended to the customers to charge the battery rack with 0.25P to certain voltage level as below:

Either a) charge to cell voltage $\geq 3.355\text{V}$,

or b) cell voltage $\geq 3.294\text{V}$ after 2 hours resting.

- Battery capacity test about 100% SOC charge and 0% SOC discharge must be conducted every 12 months
- When recover the system from idle mode over 180 days to discharge/charge mode, the battery system must perform SOC calibration applying one of the possible methods, including:
 - Static Calibration
 - Dynamic Calibration
 - High Voltage Calibration

	CAUTION
	If the energy storage system keep on idle mode for a long time, it will cause irreversible damage to the battery. Please perform regular maintenance.

9.1.3 Battery module spare parts maintenance

Spare Batteries are shipped out from CATL at SOC=30%。

- The initial maintenance of the spare battery needs to be conducted after 12 months period. Charge the battery to 50% SOC (Equipment used: Pack-level charge and discharge machine)
- After the initial recharge, charge the battery once a year. Charge the battery to 50% SOC. (Equipment used: Pack-level charge and discharge machine)
- The spare parts should be maintained as follows before use.

First discharge the battery until the minimum cell voltage is 2.8V; evaluate the battery consistency according to the OCV curve table, if the SOC imbalance (that is, the SOC corresponded to average voltage – the SOC corresponded to the lowest cell voltage)> 5%, you need to perform balanced maintenance first. (Equipment used: Pack-level charge and discharge machine、Equalization equipment)

Remark:

Charging duration (h) = (target SOC - current SOC)*nominated capacity (Ah)/machine constant charging current (A)

Discharging duration (h) = (current SOC - target SOC)*nominated capacity (Ah)/machine constant discharging current (A)

9.1.4 Battery system inspection

Conduct an inspection of ESS every 12 months and make inspection record.

9.2 Rack enclosure maintenance

	CAUTION The enclosure is not allowed to have rust, bend structure during whole product lifecycle
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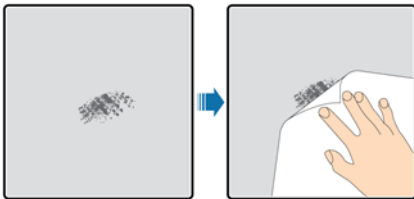
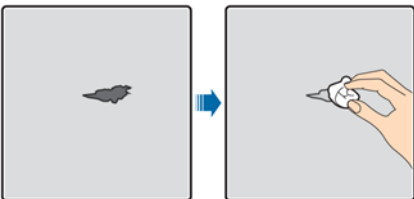
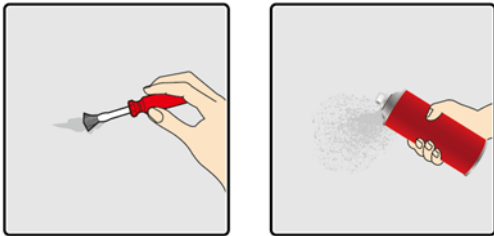
Conduct an inspection of Rack every twelve months and make inspection record, if the paint on the surface have following damage scenarios, repaint it according to the following steps.

- Shallow scratch (not exposed steel substrate)
- A stain or rust that cannot be wiped
- Deep scratch (damage of primer, exposure of steel substrate)

➤ Precondition

- In the outdoor environment without shelter, it is strictly forbidden to repaint in bad weather.
- Prepare the corresponding paint with the same color of the rack surface.
- Visually inspect the severity of the surface paint damage of the rack, prepare corresponding tools and materials, and evaluate the quantity of materials on site based on the repainting situation.
- Paint that meets requirements has been prepared based on the paint surface color number **RAL7035** of racks.
- The exterior of the rack should be intact. If the paint on the surface of the rack is damaged or broken, repaint it immediately.
- Tools and Materials: Hand spray paint or paint, brush (for painting a small area), fine sand paper, anhydrous ethanol, cotton cloth, and spray gun

➤ Procedure

Step	Procedure
1	Gently polish the damaged area with fine sandpaper to remove dirt or rust. 
2	Moisten the cotton cloth with anhydrous ethanol, wipe the polished area or area to be repaired, remove dirt and dust, and wipe dry with a clean cotton cloth. 
4	According to the damage degree of the paint, evenly apply paint to the damaged coating by using one of the following methods until no damage is visible: 1. A small number of scratches and small area of stains, rust recommended to use hand spray paint or brush paint. 2. A large number of scratches and large areas of stains and rust should be sprayed with a paint spray gun. 
5	Leave the paint on for 30 minutes, and then observe whether the repainting area meets requirements.

CAUTION	
	If the substrate is exposed in the area to be repaired, paint primer until the substrate is not exposed after the paint dries, and then apply middle layer paint and polyurethane topcoat. Use primer or polyurethane top coat of the corresponding color.

	CAUTION
	Note that the paint film should be as thin and even as possible. The paint film should not be droplet shape and the surface should be smooth. If there are different colors on the box body pattern, use adhesive tape and white paper to cover the parts of other colors except the damaged paint before painting, so as to avoid contamination of the parts of other colors during painting repair.

	CAUTION
	The paint patch area should be consistent with the color of the surrounding area, without obvious boundaries, obvious bumps, damage marks, and paint peeling phenomenon. If the user want to spray paint, it's advised to spray it three times and then check whether it meets the requirements. If not, repeat until the requirements are met.

9.3 Maintenance requirements for Thermal Management System

9.3.1 Instructions of Chiller Maintenance:

- It is strictly forbidden to let the unit work when there is no coolant in the waterway;
- It is strictly forbidden to operate the unit under the condition of soaking in water;
- It is strictly forbidden for personnel to touch the unit while the unit is running;
- It is strictly forbidden to disassemble by non-professionals
- It is strictly forbidden for the unit to be squeezed by heavy objects
- Use a vacuum cleaner to clean the condenser, radiator and air inlet and outlet of the unit every three months (no water);

9.3.2 Instructions of Coolant Maintenance:

Replace the coolant every five years or the coolant is degraded below any of the following levels.

- PH value < 6.5 or PH value > 9.5
- Chloride concentration > 60ppm
- Appearance: turbid coolant with impurities such as precipitated particles or flocculent impurities that are easily seen as shown below:

Turbid coolant:	Normal coolant:
	

If coolant has not degraded below any of the conditions described above by YAE, existing coolant can be used until the next scheduled inspection (every 5 years).

The coolant element is EC-45, only following approved supplier & type can be used

Table 9-2: Coolant element

Coolant	Supplier	Supplier type	Mark
EC-45	BASF	G30-45	
	SINOPEC	CNSH OEVC-45	
	Lopal Tech	JSLP L70-45	Default type

	CAUTION Mixed coolant from different supplier or type is not allowed, otherwise, the cooling capacity will be lost.
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	CAUTION Any damage to the relevant parts of the battery compartment caused by using other coolants which are not recommended or approved by CATL is not covered by the warranty.
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9.4 Spare Parts

The detail spare parts list can be provided order by order, the range of spare parts is pre-defined as following:

- Control box and its sub components
- Battery module and CSC module
- Fire Suppression System components
- Thermal Management System components
- Enclosure

10 System Decommissioning & Removal

Contact CATL Service before attempting to decommission or remove a system. CATL equipment is designed for end-of-life management through recycling and materials reclamation, tasks which are normally handled by CATL-qualified partners. CATL Energy Storage can supply a list of qualified partners in your area to help ensure proper disposal of retired equipment.

11 Emergency Plan

After anomalies and accidents occur in battery system, correct and effective treatment measures shall be taken in time to deal with the problems in order to eliminate further damage and enlarged losses. The detail emergency plan can be separated provided.